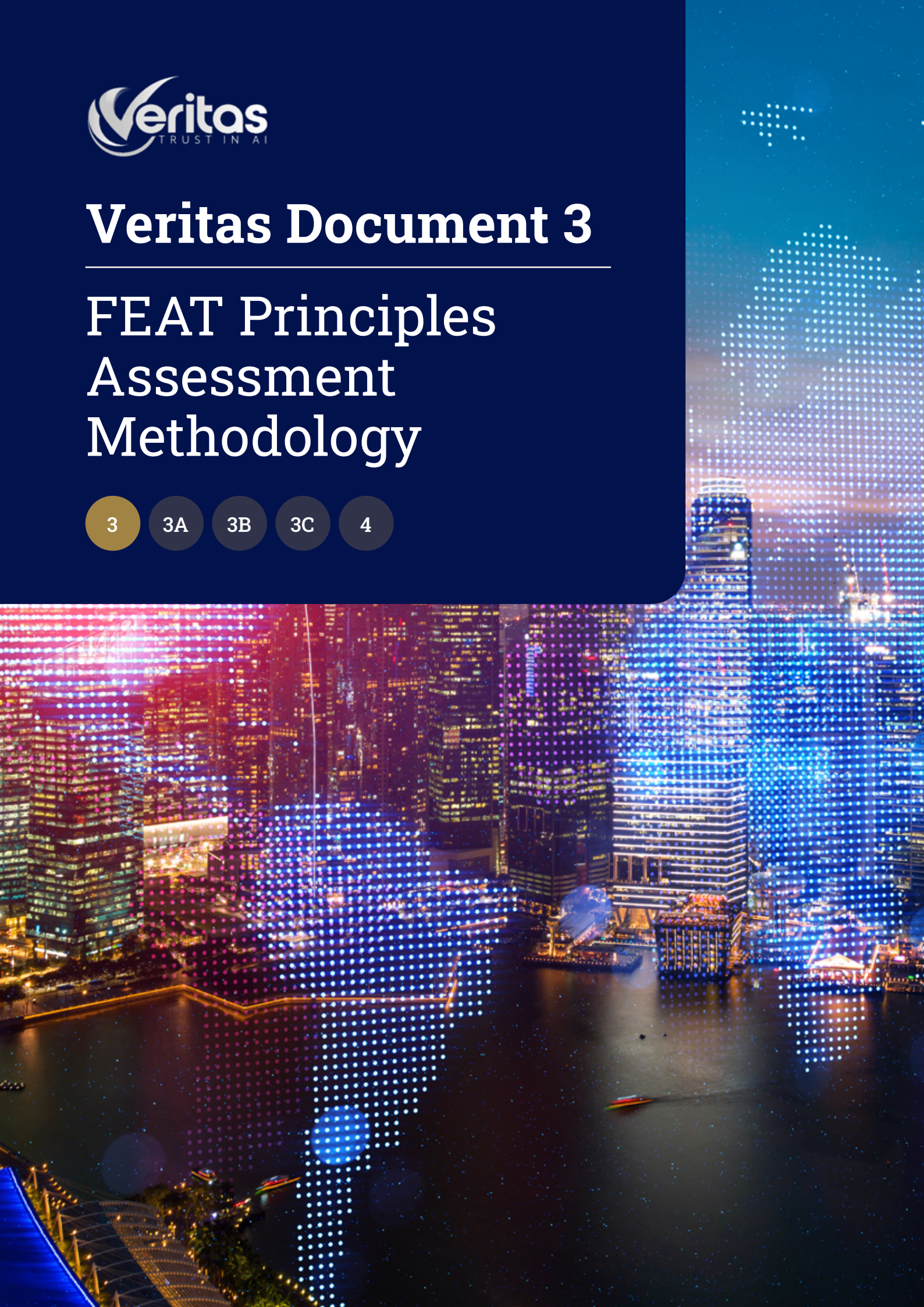
# Veritas Document 3

## FEAT Principles Assessment Methodology

3 | 3A | 3B | 3C | 4



# Foreword

The Veritas consortium, comprising the Monetary Authority of Singapore (MAS) and industry partners, is pleased to present the Veritas* Phase Two white papers which assessed the alignment of the AIDA systems against the fairness, ethics, accountability and transparency (FEAT) principles. These white papers contain the collective experience of the Veritas consortium and share the generic FEAT Assessment Methodology. They also include the illustrations for assessing the FEAT principles for selected use cases related to predictive underwriting, customer marketing, fraud detection and credit risk scoring.

In January 2021, MAS published two documents as part of Veritas Phase One - FEAT Fairness Principles Assessment Methodology and FEAT Fairness Principles Assessment Case Studies. The focus for Veritas Phase One was on the development of the Fairness Assessment Methodology in credit risk scoring and customer marketing for the banking industry. In Phase Two, the methodology was extended to cover all FEAT principles and will be applicable for more use cases. The white papers also highlight the existing global regulatory landscape and how it provides guidance to the Responsible AI adoption by financial institutions.

With the conclusion of Phase Two, the Veritas initiative has made significant progress in providing a full FEAT Assessment Methodology. This Assessment Methodology will enable financial institutions to assess the alignment of their AIDA systems against FEAT principle systematically. In addition, the Veritas consortium has also developed an open source Veritas Toolkit that implements the Veritas FEAT Assessment Methodology. The first version of Veritas Toolkit was released together with the Phase 2 white papers and the toolkit utilises the best fairness tools on the market to help financial institutions in Singapore to begin adoption of Veritas Fairness Assessment Methodology in their own organisation.

For the next phase of Veritas, the consortium will continue to enhance Veritas FEAT Assessment Methodology and apply it to additional use cases in the financial services sector. The consortium will also work on a Version 2 of the Veritas Toolkit, which will provide full FEAT assessment functionality.

We encourage financial institutions and tech firms to read these white papers and implement the methodology in their own organisations, as well as to utilise the Veritas Toolkit. The Veritas consortium welcomes feedback so that it can make further improvements. Let us work together to ensure that AIDA is used in a more responsible manner in Singapore's financial sector.

We would like to express our appreciation to all the members of the consortium for their active participation and generous support in the development of these white papers. We would also like to acknowledge the contributions of our industry partners - Accenture, AXA, HSBC, Standard Chartered, Swiss Re, TruEra and UOB in this remarkable endeavour.



**Sopnendu Mohanty**
**(MAS CFTO)**

---

**Veritas was announced as part of Singapore's National AI Strategy by Mr Heng Swee Keat, Deputy Prime Minister of Singapore, at the Singapore FinTech Festival and Singapore Week of Innovation and TeCHnology (SFF x SWITCH) 2019.*

# Table of Contents


**Executive Summary** ..... 5

**1 Introduction** ..... 6

1.1 Background of Project Veritas Phase 2 6

1.2 Structure of Veritas Phase 2 Methodology Documents 8

1.3 Audience of Veritas Phase 2 Methodology Documents 9

**2 FEAT Principles Assessment Methodology** ..... 10

2.1 Context 10

2.2 Navigating this Document 10

2.3 Designing AIDA Systems for FEAT Consideration 11

2.4 Recap of the FEAT Principles 12

2.5 Interdependencies Across the FEAT Principles 13

**3 Application of FEAT for the Development Lifecycle of an AIDA System** ..... 15

3.1 Embedding the FEAT Principles Assessment Methodology into the Development Lifecycle of an AIDA system 15

3.2 Foundational Setup 18

**4 Summary of FEAT Principles Assessment Methodology** ..... 25

4.1 Fairness 25

4.2 Ethics and Accountability 26

4.3 Transparency 27

# Table of Contents


**5 Global Regulatory Landscape on Regulating Responsible AI** ..... 29

5.1 Global Regulatory Landscape 29

5.2 Key Regulatory Trends 30

5.3 Assessing the Regulatory Implications of AIDA Projects in Financial Services 30

5.4 Adopting a Materiality Based Approach to Assessing Regulatory Considerations 31

5.5 Leverage Existing Risk Management and Governance Frameworks 31

5.6 Collaborate with Technology Partners 32

5.7 Ensure a Diversity of Perspectives 33

5.8 Share Use Cases with the Industry 33

5.9 The Way Forward for Veritas 33

**6 Acknowledgements** ..... 34

**7 Bibliography** ..... 35

**8 Appendix A** ..... 41

**9 Appendix B** ..... 46

**10 Appendix C** ..... 50

**11 Annexure** ..... 58

# Executive Summary

In 2018, The Monetary Authority of Singapore (MAS) released Principles to promote Fairness, Ethics, Accountability and Transparency (FEAT) in the use of Artificial Intelligence and Data Analytics (AIDA) in Singapore's Financial Sector. The Veritas initiative was subsequently commenced in 2019 as part of Singapore's National AI Strategy.

Veritas is a multi-phased collaborative project with financial industry. In January 2021, MAS announced the successful conclusion of the first phase of the Veritas initiative which focused on the development of the fairness assessment methodology in credit risk scoring and customer marketing for the banking industry. The Veritas Consortium, comprising MAS and industry partners, published whitepapers on the fairness assessment methodology and the open source code of these two use cases.

Veritas Phase 2 extends the focus to cover all FEAT Principles ("Principles") for its adoption by FSIs beyond banks to include coverage of the insurance industry. In Veritas Phase 2, the consortium leveraged the Methodology from phase 1 as a foundational baseline to evolve and provide integrated guidance across the FEAT Principles. This FEAT Principles Assessment Methodology whitepaper is one of a suite published as a deliverable of the Veritas Phase 2 project.

This whitepaper is an overarching document that provides a holistic view of the application of FEAT Principles to AIDA system development lifecycles. As the foundational steps, this whitepaper provides a list of recommendations for FIs on how to:

- Bring a FEAT assessment to scale
- Define materiality and risk tiers
- Adopt a risk-based governance approach for application of FEAT Principles to use cases
- Integrate with existing risk management practices

FEAT Principles should be incorporated at an organisational level down to use case or AIDA system level, where relevant. FEAT Principles are highly interdependent and must be mapped across the AIDA system development lifecycle for them to be operationalised. A checklist is provided across the end-to-end AIDA lifecycle as well as considerations to assist FSIs in responding to assessment questions. While most of the questions in the checklist are aligned to the individual FEAT Principles, some are general questions that will more broadly help organisations assess their AIDA systems. It is recommended to start with Ethics and Accountability assessment which helps to define the guiding principles at an organizational level, then perform the Fairness and Transparency assessments concurrently within the various steps of the AIDA system development lifecycle, as they are inter-related and are use case specific.

This overarching document also provides a summary of the deep dive Methodology whitepapers and a review of the global regulatory landscape around Responsible AI adoption.

FSIs are expected to work towards achieving the FEAT Principles and outcomes as they increasingly deploy AIDA to make decisions. FSIs, however, have flexibility to explore how best to achieve them, including establishing governance frameworks and control processes that are appropriate to their business operations and risk profiles.

The Phase 2 documents do not prescribe what FSIs must do to comply with FEAT, but provide practical and detailed guidance on how FSIs can perform FEAT assessment of their AIDA models and systems, which FSIs could use and adapt as appropriate.

# 01 Introduction

## 1.1 Background of Project Veritas Phase 2

In 2018, The Monetary Authority of Singapore (MAS) released Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the use of Artificial Intelligence and Data Analytics (AIDA) in Singapore's Financial Sector. The Veritas initiative was subsequently commenced in 2019 as part of Singapore's National AI Strategy, as an industry wide collaboration to provide guidelines to Financial Services Institutions (FSIs) to incorporate the FEAT Principles into AIDA solutions.

Veritas Phase 1, which concluded in January 2021, focused on the development of the Fairness Principles Assessment Methodology in credit risk scoring and customer marketing for the banking industry. MAS published two documents as part of Veritas Phase 1: FEAT Fairness Principles Assessment Methodology[1] and FEAT Fairness Principles Assessment Case Studies.[2]

Veritas Phase 2, which started in March 2021, extends the focus to cover all FEAT Principles ("Principles") for its adoption by FSIs beyond banks to include coverage of the insurance industry. In Veritas Phase 2, the consortium leveraged the Methodology from phase 1 as a foundational baseline to evolve and provide integrated guidance across the FEAT Principles.

Veritas Phase 2 was commenced with the below objectives:

- Translate and operationalise the FEAT Principles.
- Extend the Phase 1 Methodology (Fairness) beyond banks to include coverage of the insurance industry.
- Enable FSI partners with FEAT adoption through Veritas Framework with solution prototypes (i.e., guidelines, codes, visualisation, etc.) and use case illustrations.

The consortium of Veritas Phase 2 was also expected to:

- Streamline collaboration and methodology across the FEAT Principles.
- Share industry expertise through the involvement of academics, industry researchers and practitioners.
- Promote industry adoption of the FEAT Principles Assessment Methodology through workable prototypes.

Specifically, Veritas Phase 2 comprised three core working groups to develop the assessment methodology for their focus areas. Information about the working members in the group can be found below (Figure 1.1).

| | Lead Members | Use Cases |
|---|---|---|
| **Fairness** | Swiss Re, accenture | Insurance Underwriting |
| **Ethics & Accountability** | AXA, UOB, accenture | • Customer marketing (Banking)<br>• Fraud detection (Insurance) |
| **Transparency** | truera, HSBC, standard chartered | • Credit risk scoring (Banking)<br>• Customer marketing (Banking) |

*Figure 1.1 Core working group members and their workstreams*

Veritas Phase 2 had a total of 27 members in the consortium (as shown in Figure 1.2) responsible for leading and supporting the effort of developing the FEAT Principles Assessment Methodologies.

| Lead Members | Members (FSIs) | Facilitators | Advisory Members |
|---|---|---|---|
| **Drive the Veritas development of the user cases** | **Provide domain knowledge & feedback on the use cases** | **Facilitate the whole program** | **Provide expertise & review the whitepaper** |
| HSBC, Swiss Re, standard chartered, UOB, truera, AXA, accenture | ICBC 中国工商银行, DBS, citi, BNP PARIBAS, PING AN, 中国银行, MUFG, Goldman Sachs, UnionBank, OCBC Bank, PRUDENTIAL, BNY MELLON, BANK OF SINGAPORE | accenture, MAS Monetary Authority of Singapore | Microsoft, BAFT, aws, CFA Society Singapore, Google, IBM |

*Figure 1.2 List of Veritas Phase 2 consortium members*

## 1.2 Structure of Veritas Phase 2 Methodology Documents

In Phase 2, which spans four FEAT Principles and three working groups, the documents are structured into methodology and use case related whitepapers. This FEAT Principles Assessment Methodology whitepaper is an overarching document that provides:

1. A holistic view of the application of FEAT Principles to AIDA system development lifecycle.
2. A summary of the deep dive Methodology whitepapers and a review of the global regulatory landscape around Responsible AI adoption.

Figure 1.3 provides a quick overview of the documents that will be published as part of Veritas Phase 2. The documents detailing the Methodology for each principle should not be viewed as standalone documents, but instead as parts of an end-to-end Methodology across an AIDA system.

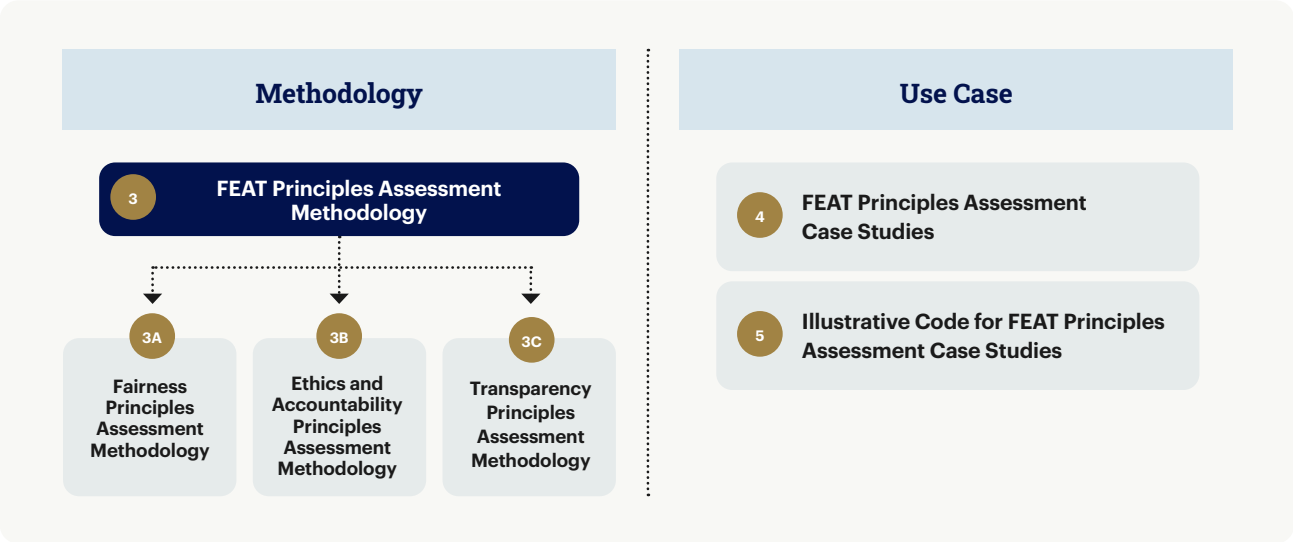



*Figure 1.3 List of documents in Veritas Phase 2*

## 1.3 Audience of Veritas Phase 2 Documents

Like the Veritas Phase 1 Methodology, this document distinguishes between four groups involved in AIDA system assessments in different ways.

**AIDA system developer:** An AIDA system developer is the party that develops the AIDA system under study. This could be an FSI or a vendor or a Fintech firm, for example.

**AIDA system owner:** An AIDA system owner is the party using the AIDA system to help make decisions for fulfilling their business goals. These decisions have fairness and/or ethical implications for the customer or other affected parties or the organisation. This role could be played by the business team in the FSI that is responsible for the AIDA system.

**AIDA system assessor:** An AIDA system assessor is the party that assesses the FEAT assessment presented to them by FSI. This role could be played by an external audit firm, or internally by a model risk management or internal audit group.

**FSI:** An FSI is the organisation responsible for the AIDA system under study. The AIDA system owner sits within the FSI and is acting on its behalf. The AIDA system developer may be part of the FSI if the system is developed in house or may be an organisation external to it if the system has been procured. Similarly, the AIDA system assessor may sit within the FSI (for an internal assessment) or be a separate body (for an external assessment).

# 02 FEAT Principles Assessment Methodology

## 2.1 Context

AI has become a top technology priority among FSIs in recent years.[3] According to a 2021 Gartner CIO Survey:

- 32% of banking respondents and 36% of insurance respondents expect AI to be a "game changer" for their enterprise in 2021.
- Approximately 25% of banks and 31% of insurers have already deployed AIDA in 2020 and an additional 25% of banks and 23% of insurers plan to deploy AIDA in the next 12 months.

Significantly, however, according to another global executive survey, while 60% of respondents believe that the adoption of AI is necessary to have a competitive advantage in a digital economy; 45% agree that not enough is understood about the unintended consequences of AI.[4]

Given the increasing adoption of AIDA within FSIs, the industry's regulators may work to ensure compliant and ethical use of the technology to prevent potential negative outcomes or harm as defined in the FEAT Principles. Recognising this opportunity to mitigate potential harms and unintended consequences of AIDA, MAS had collaborated with the financial industry and other relevant stakeholders to create a set of Principles guiding the responsible use of AIDA in financial services.

## 2.2 Navigating this Document

The structure of this document is as follows:

- **Section 3** outlines the overall FEAT considerations for designing AIDA systems in FSIs, the FEAT Principles, how they map to each of FEAT areas, and interdependencies across the Principles. Section 3 also explains how to embed the FEAT Principles Assessment Methodology to an AIDA system's development lifecycle and gives guidance on foundational steps, how to adopt a risk based approach and how to scale FEAT across an organisation.
- **Section 4** provides a summary of three working groups' Methodology Frameworks.
- **Section 5** includes information about the global regulatory landscape for the responsible adoption of AIDA within the financial sector.
- Finally, the **Appendices** include a checklist for assessing adherence to FEAT guidelines, a glossary with common terminology used across the FEAT documents, and a section on the global AI regulatory landscape.

## 2.3 Designing AIDA Systems for FEAT Consideration

Artificial Intelligence and Data Analytics comprises technologies that assist or replace human decision making. In the last decade, the increased availability of data from digital channels and third party sources has allowed use of AIDA by financial institutions to evolve widely, enabling organisations to perform quantitative risk aware decision making across functions such as business operations, customer service and marketing to achieve more sophisticated predictions, personalisation and insights, risk management, and compliance at a vast scale.

An AIDA system contains one or more AIDA models that work together to achieve a set of objectives. The decisions made by an AIDA system are "AIDA driven decisions". AIDA systems encompass not just the AIDA model(s), but also other parts of a decision making process that are required to get to the final AIDA driven decision such as:

- Human contributions that affect AIDA decisions (humans might give input or use the model output as a factor to inform their final decision).
- Other governance or rules that affect decisions.
- Actions taken based on the AIDA system's output.

Monitoring and review mechanisms are also part of the end-to-end AIDA system and will impact AIDA driven decisions over time as feedback from monitoring and review is used to update elements such as the model, business rules and human inputs, for example.

AIDA systems developed, deployed and monitored using the FEAT Principles Assessment Methodology can minimise human bias, which is manifestly beneficial to consumers. To evaluate the application of the FEAT Principles Assessment Methodology to an AIDA system, financial institutions need to:

- Define overall FEAT Principles that are aligned to their organisation's values and principles.
- Assess, measure and monitor how the AIDA system achieves these Principles.
- Regularly assess overall commitments and refine the Principles.

In Phase 2, the consortium reviewed and assessed all four Principles of: Fairness, Ethics, Accountability and Transparency, as well as evaluated these Principles against a set of use cases. The findings conclude that FEAT Principles and considerations can exist at both an AIDA system/use case level and at an overall organisational level, and that these may differ according to geographical location and societal norms. Regulated entities are encouraged to incorporate FEAT Principles at an organisational level down to use case/AIDA system level, where relevant.

FEAT Principles are highly interdependent and must be mapped across the AIDA system development lifecycle for them to be operationalised. While all FEAT Principles are to be applied at an organisation level, the Fairness and Transparency Principles are tailored to the use case or specific system (this can be seen in Veritas Phase 1, where the fairness assessment for customer marketing was significantly different to that for credit scoring). Ethics and Accountability Principles focus on defining clear and consistent organisational values and principles, which sets the foundation for translating Principles to practice by introducing commitments and specifications to operationalise values for each AIDA system or use case. While the assessment of ethics and accountability happens at the foundation stage, elements of the Ethics and Accountability Methodology are applicable throughout the use case lifecycle.

## 2.4 Recap of the FEAT Principles

MAS released a set of 14 Principles that cover Fairness, Ethics, Accountability and Transparency for responsible AIDA adoption within FSIs[5]. The following figure 2.1 shows how the Principles are mapped to each of the FEAT areas.

| | |
|---|---|
| **Fairness** | • **P1:** Individuals or groups of individuals are not systematically disadvantaged through AIDA-driven decisions, unless these decisions can be justified.<br>• **P2:** Use of personal attributes as input factors for AIDA-driven decisions is justified.<br>• **P3:** Data and models used for AIDA-driven decisions are regularly reviewed and validated for accuracy and relevance, and to minimise unintentional bias.<br>• **P4:** AIDA-driven decisions are regularly reviewed so that models behave as designed and intended. |
| **Ethics** | • **P5:** Use of AIDA is aligned with the firm's ethical standards, values and codes of conduct.<br>• **P6:** AIDA-driven decisions are held to at least the same ethical standards as human-driven decisions. |
| **Accountability** | • **P7:** Use of AIDA in AIDA-driven decision-making is approved by an appropriate internal authority.<br>• **P8:** Firms using AIDA are accountable for both internally developed and externally sourced AIDA models.<br>• **P9:** Firms using AIDA proactively raise management and Board awareness of their use of AIDA.<br>• **P10:** Data subjects are provided with channels to enquire about, submit appeals for and request reviews of AIDA-driven decisions that affect them.<br>• **P11:** Verified and relevant supplementary data provided by data subjects are taken into account when performing a review of AIDA-driven decisions. |
| **Transparency** | • **P12:** To increase public confidence, use of AIDA is proactively disclosed to data subjects as part of general communication.<br>• **P13:** Data subjects are provided, upon request, clear explanations on what data is used to make AIDA-driven decisions about the data subject and how the data affects the decision.<br>• **P14:** Data subjects are provided, upon request, clear explanations on the consequences that AIDA-driven decisions may have on them. |

*Figure 2.1 FEAT areas mapped to 14 Principles*

## 2.5 Interdependencies Across the FEAT Principles

FSIs need to focus on assessing AIDA systems from Fairness, Ethics, Accountability and Transparency perspectives as a whole to meet Responsible AI adoption obligations. As some of the FEAT Principles are interdependent, FSIs should consider the below when assessing the Principles.

- **Dependency of Fairness Principles on other FEAT Principles.** P1-P4 discusses fairness aspects such as the justification of systematic disadvantage of individuals or groups, the use of personal attributes, regular reviews of data and models, and the decisions that FSIs need to take when using AIDA systems for decision making. These principles are dependent on:
  - **Ethics Principles 5 and 6.** The use of AIDA to align with the ethical values of the organisation, to reflect on a choice of which (if any) personal attributes to be excluded when developing AIDA models, and regular reviews and monitoring of models used in decision making.
  - **Internal Accountability Principles 7 and 8.** Approval of internal stakeholders for AIDA driven decision making is necessary, as some may have to take responsibility for those decisions. Regular reviews of data and the model could support a firm's accountability for internal/external sourced models.
  - **External Accountability Principle 10.** FSIs are encouraged to have channel for appeals/redress for customers to submit review request of an AIDA decision that has disadvantaged them. Mechanisms for appeals, also provide a channel for FSI to redress/mitigate systematic harms.
  - **Transparency Principles 13 and 14.** Regular review of internal/external sourced models based on fairness principles and the explanation of how it arrived at certain decisions could help to support firm's management / board accountability obligations. In addition, customer facing transparencyenables firms to meet the requirements for providing customers with a channel for redress where necessary.
- **Dependency of Ethics and Accountability Principles on other FEAT Principles.** P5-P11 focus on the ethics and accountability aspects covering ethical standards, the awareness of internal stakeholders and management with regard to use of AIDA systems, and the need to provide a necessary channel for redress. Dependent Principles include:
  - **Transparency Principles 13 and 14.** Firms' management/board can only fulfil their accountability obligations if they can demonstrate that they, or people working on their behalf, are able to explain how an AIDA system reached a given outcome. To that extent, internal transparency is essential. Meanwhile, customer facing transparency enables firms to meet the requirements for providing customers with a channel for redress where necessary.

- **Dependency of Transparency Principles on other FEAT Principles.** P12-P14 covers both internal and external transparency aspects of the FEAT Principles. Dependencies on other Principles include:
    - **Ethics Principles 5.** Decisions on the level of transparency that customers will be provided with will be a function of the firm's ethical standards, values and code of conduct (unless there is explicit regulatory requirements from either the financial services regulator or the data regulator) along with other considerations around areas including fraud and anti-money laundering.
    - **External Accountability Principle 10.** Providing a channel to customers to ask questions (and seek redress), submit appeals for and request reviews is essential for P13 and P14 (both of which are "upon request") to be possible where applicable.

# 03 Application of FEAT for the Development Lifecycle of an AIDA system

## 3.1 Embedding the FEAT Principles Assessment Methodology into the Development Lifecycle of an AIDA system

To perform a holistic assessment of the FEAT Principles, FSIs should embed the Methodology into their AIDA system development lifecycle. As a prerequisite step, it is recommended that organisations set out guiding principles with regard to developing AIDA systems in line with their corporate values. These values are defined in line with the Ethics and Accountability Methodology at an organisational level (or at the business unit level for larger companies with different business segments serving different business objectives). To put these principles into practice, it is important to translate them into practical steps supported by metrics and thresholds and embedded into everyday practices throughout the AIDA system lifecycle.

As discussed, Fairness and Transparency Principles are inter-related and are use case specific. Therefore, it is important for Fairness and Transparency Methodologies to be incorporated within the various steps of the AIDA system development lifecycle. It is recommended that FSIs perform the Fairness and Transparency assessments concurrently. For example, in a customer marketing use case, FSIs will need to identify protected features from other features used in developing AIDA models to ensure fairness – they will therefore require transparency around which features are being used.

The FEAT Fairness Principles Assessment Methodology established in Phase 1 sets out five parts (A-E), which include a set of checklist questions to guide organisations. In Phase 2, the five parts methodology is used as a baseline mapped to a typical AIDA system development lifecycle and revised with new considerations around Ethics, Accountability and Transparency.

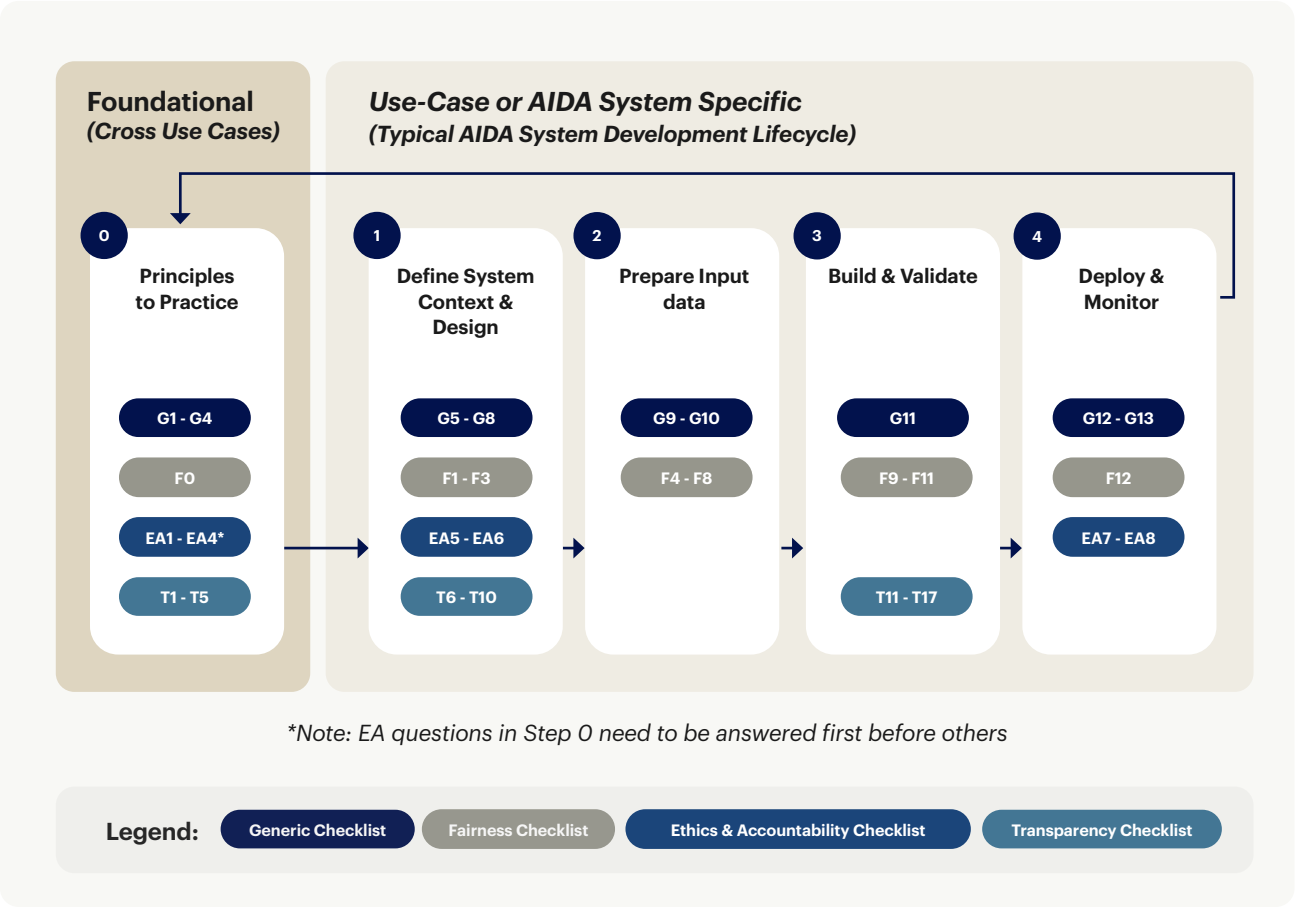



*Figure 3.1 FEAT assessment checklist mapped against the AIDA system development lifecycle*

As part of Phase 2, the FEAT Assessment Methodology documents provides a checklist across the end-to-end AIDA lifecycle (as shown in the figure 3.1) as well as considerations to assist FSIs in responding to assessment questions (to view the checklist, see appendix A). While most of the questions in the checklist are aligned to the individual FEAT Principles, some are general questions that will more broadly help organisations assess their AIDA systems. Answers to these questions should have already been available in existing AIDA system development documentation, as they are not specific to the FEAT assessment.

The general questions are foundational in nature and can be split across the five steps of AIDA lifecycle:

### 0 Step 0 – Principles to Practice

- **G1** - Is AIDA defined in the FSI?
- **G2** - Is there a framework in place to define roles and responsibilities for AIDA projects/use cases?
- **G3** - Is there a materiality framework in place? If so, do you have a clear definition to determine the different level of materiality?
- **G4** - Does the FSI have an up-to-date inventory that tracks all material AIDA use cases in use?

### 1 Step 1 - Defining System Context and Design

- **G5** - Have you documented the commercial objectives of the system and the quantitative measures to meet those commercial objectives? Is it documented how is AIDA used to achieve them?
- **G6** - Are there scope boundaries of the proposed system? For example, specific geographies/ segments?
- **G7** - Is AIDA use justified for this particular objective/use case (based on limitations of alternatives, proof of AIDA being used successfully elsewhere, availability of adequate data to train the AIDA system, valid legal basis, etc.)?
- **G8** - Has the materiality of the use case been determined using the framework defined in Step 0?

### 2 Step 2 - Prepare and Input Data

- **G9** - Have you documented the data used by the system including types of data available, the provenance (source), attributes, dictionary (meaning), vintage (time period), data (quality) profile, representativeness and informed consent? For any personal data used by the system, has a DPIA been conducted?
- **G10** - Have you documented the data pre-processing and engineering done by the system?

### 3 Step 3 - Build and Validate

- **G11** - Is the composition of the AIDA System defined? Is it clear how each component of the system AIDA models, as well as business rules and human judgement if relevant, are used to achieve its commercial objectives? Are the performance estimates and the uncertainties of those estimates documented?

### 4 Step 4 - Deploy and Monitor

- **G12** - Is the system's monitoring and review regime designed to detect abnormal operation (e.g., significant model and/or data drift)?
- **G13** - Is there fallback and/or mitigation plans in place in case of triggers from the system's monitoring and review regime?

Importantly, FSIs should always adhere to all relevant legal and regulatory requirements when undertaking activities relating to the collection, use, processing or disclosure of personal attributes involving personal data. This can be achieved by conducting a Data Protection Impact Assessment (DPIA) prior to reviewing the FEAT checklist. A DPIA involves identifying, assessing and addressing personal data protection risks based on the organisation's functions, needs and processes.

The assumption in this document is that all other relevant legal and regulatory requirements and best practices are being followed with respect to AIDA systems including but not limited to, anti-trust/competition law, security and sustainability (or ESG) considerations and other areas that fall outside the scope of this document.

Finally, when conducting the FEAT checklist review, FSIs are also encouraged to document all the decisions made and measures they put in place.

## 3.2 Foundational Setup

As indicated by the question G1 - G4 above, "Principles to Practice" is a foundational step of the AIDA system development lifecycle, preparing the way for the design, build and deployment of the system. This phase is important to ensure that an organisation's values are effectively operationalised in their AIDA systems, and is achieved by establishing a framework of governance principles to continuously track, monitor, and enhance how values are embedded into the AIDA model lifecycle.

Questions G1 - G4 in the FEAT checklist relate to the foundational stage and can be used by FSIs to guide their work in creating a governance framework anchored in their core corporate values, ethical guardrails, and accountability frameworks.

It is important that these values and principles are translated into specific AIDA system related decisions and metrics that can be tracked over a period of time and which include clearly assigned ownership and accountability. It is equally important for stakeholders across specific AIDA use cases to have visibility of the governance and metrics being tracked to ensure compliance with relevant regulations and internal standards.

There are several other key considerations for FSIs to work through during the foundational phase. The subsections below provide a list of recommendations for how to:

- Bring a FEAT assessment to scale (section 3.2.1).
- Define materiality and risk tiers (section 3.2.2).
- Incorporate a risk-based governance approach for application of FEAT Principles to use cases (section 3.2.3).
- Integrate with existing risk management practices (section 3.2.4).

Establishing good governance principles is important for all firms, regardless of size. All firms are encouraged to focus on implementing the AIDA system on areas where they have sufficient resources to undertake a FEAT assessment to the level of rigor required for the risk level posed by the AIDA system.

### 3.2.1 Creating the Right Foundation for Scaling FEAT Across an Organisation

FSIs need to ensure that their FEAT assessments can be scaled to AIDA usage across the organisation in a consistent and robust way. The following set of recommendations are intended to provide guidance to the FSIs in this regard. The below points are key to the overall success of embedding FEAT principles into day-to-day processes of an FSI.

- **Employ a materiality assessment framework.** Firms can use the assessment framework outlined in this paper (see section 3.2.2 for more details) or use existing risk assessment frameworks to tier the risk of AIDA systems and define the governance policies required to meet different business needs. The goal of risk tiering is to ensure that systems with higher levels of risk are prioritised in terms of focus and resources.
- **Embed FEAT related considerations into existing risk frameworks and governance structures.** Frameworks around model risk (or its equivalent outside banking), cybersecurity, data privacy and the fair treatment of customers can be adapted for FEAT considerations. Ensure relevant and transparent cross domain governance structures are updated to include FEAT Principles, including identifying roles, expectations, and accountability. This step should be undertaken prior to Stage 1 ("Defining System Context and Design") in the AIDA system development lifecycle to inform the design of the AIDA system (see section 3.2.4 for more details).
- **Define standards and processes.** Definitions should cover all key areas to bring consistency and clarity to the processes. Organisations should look to define the right metrics, specifications, and commitments to deliver the consistent scaling of FEAT assessment process across the enterprise.
- **Invest in appropriate training.** Training programmes to include customers, staff, management, and risk/control functions and seek to demystify AIDA systems to increase trust in their outcomes. Organisations should also identify new and changing roles and map areas where they need to upskill, reskill or hire to accommodate new ways of working and enable effective human + machine collaboration.
- **Do not address any of the FEAT principles in isolation.** FEAT Principles are recommended to be embedded throughout the AIDA system lifecycle, and no one Principle to be treated as a one off or standalone task with regard to model development, testing and/or validation.
- **Articulate your key principles internally and, if possible, externally.** FSIs need to communicate a proactive, consistent message on their use of data and algorithms and the impact of AIDA decisions to customers, employees and other stakeholders. This could be conducted through websites and/or other electronic and physical channels. Communications should incorporate the company's Responsible AI mission, including its values around ethics, accountability, transparency and fairness, and be informed by brand as well as public risk assessments and guidance.
- **Create and encourage an organisational culture that empowers individuals to raise doubts or concerns about AIDA systems.** Organisations should create a culture that incentivises the application of the FEAT Principles and which holds AIDA system owners accountable for achieving Responsible AI goals. Only then will the organisation begin to assess every AIDA system through a responsibility lens and start thinking more deeply about how to track relevant data and measure change.

- **Supplement use case level investments in tools for AIDA systems to embed FEAT across use cases.** This approach delivers more systemic support to teams across the organisation to help them meet their AIDA obligations. Examples of such capabilities include a single customer level dashboard that provides an overview of data held by the FSI about data subjects and where their data is used within the FSI, as well as a centralised AIDA model repository.
- **Incorporate insights from multidisciplinary teams into the FEAT assessment process and ensure sponsorship from the leadership.** This includes talent specialised in areas such as human resources, behavioural sciences, law, human centric design etc., to provide the right direction to AIDA based programmes. The better the blend of skills, the more sustainable the result and the better the governance.

### 3.2.2 Determining the Materiality of AIDA Systems

One of the key elements in defining an AIDA governance model is to assess the materiality of each AIDA use case. The first step in managing risk is to determine the risk level posed by the AIDA system and its impact on the organisation. Traditionally, model risk identification and tiering have given more weight to model impact (materiality and criticality) considerations. Owing to the differential risk attributable to certain use cases and the AIDA systems which support them, it has been observed that models can give rise to unintended consequences. These include bias towards certain groups even when sensitive characteristics are excluded from models, or the inappropriate projection of a model to populations for which it has not been trained.

The FEAT Principles acknowledge that their application should be scaled according to the risk of a particular AIDA system, stating that "firms can calibrate actions and requirements under their internal governance framework based on the materiality of the AIDA driven decisions".[6]

As mentioned, the risk assessment of an AIDA system results in risk tiers, which determine the subsequent governance and design of the AIDA systems. These tiers should enable the FSIs to manage risk by focusing resources on high risk AIDA systems with proper guardrails and mitigating actions.

Phase 1 defined the parameters to identify risk as follows:

- **Extent to which AIDA is used in decision-making.** More complete substitution of technology for human decision-making may increase the risk of accidental systematic disadvantage, especially if the AIDA models are more complex and opaque.
- **Extent of automation of process of AIDA-driven decision-making.** All else being equal, more automation is likely to mean a higher volume and/or speed of AIDA-driven decision-making which may increase the scope of impact and decrease opportunities for timely human intervention to prevent issues.
- **Complexity of the AIDA model.** This risk relates to the complexity of the mathematical computation, the number and type of data sources used as inputs, or the level of human intervention in addition to autonomous decision making. More complex models may result in a model with limited interpretability (i.e., they could be "black boxed") or they may increase the risk that the use of particular attributes is not fully justified.

- **Severity and probability of impact on different stakeholders, including individuals.** For example, a use case for credit approval or pricing decisions is of much greater consequence to individuals than, for instance, that for new marketing messages. The impact could also include the volume and scope of usage, and the type and number of external and internal stakeholders that can be affected.
- **Monetary and financial impact.** AIDA models may be higher risk because of the potential impact to the FSI, such as its exposure and balance sheet, or to customers.
- **Regulatory impact.** Some AIDA models may be higher risk because of the regulatory exposure of their decision making for applications such as anti-money laundering.
- **Options for recourse.** All else being equal, if there are efficient and effective ways for customers or other stakeholders to challenge an AIDA system's decisions, then the associated risks (such as systematic disadvantage) may be lower.

In addition to the above, FSIs can also include impacts related to reputational risk, and the use of personal data in AIDA systems:

- **Reputational risk.** The decisions resulting from some AIDA models may have a higher risk potential for negative publicity, public perception or uncontrollable events that have an adverse impact on a company's reputation.
- **Use of personal data.** The extent to which justifiable personal data can be used for certain use cases such as the use of personal medical history for health insurance policies.

With the above in mind and by way of illustration, credit scoring with auto decline would be a high risk use case, because of the model complexity, level of automation and potential impact on individuals. Using the parameters above, firms can judge the risk profile of other use cases.



### 3.2.3 Incorporating a Risk Based Governance Approach to the Application of the FEAT Principles

FSIs are encouraged to customise the depth of the FEAT Principles Assessment Methodology to be commensurate with the risk level and the subsequent risk tiering identified for the AIDA system under consideration. Such customisation could relate to the assessment process itself, the level of detail of the assessment, or the parties involved. FSIs should ensure that customisation aligns with their existing practices to ensure consistency across the firm.

Dimensions for customisation of the assessment methodology include:

- **Extent of analysis and level of documentation:**
  - For higher risk use cases, the assessment would require more extensive analysis and more thorough responses to the questions laid out in the FEAT Principles Assessment Methodology.
  - For lower risk use cases, it may be sufficient to provide short summary answers to some or all of the assessment questions.
- **Trade-offs between business and FEAT objectives:**
  - Higher risk use cases may call for greater efforts towards upholding FEAT Principles, at the expense of business objectives.
  - In such situations, other means of improving the business performance of systems, such as improving data collection volume or quality, may be explored to achieve a more optimal balance.
- **Frequency and extensiveness of monitoring regime:**
  - Higher risk use cases are likely to require more frequent and in depth monitoring processes to be put in place following deployment.
  - Risk appropriate decisions should be made regarding the metrics and outcomes to be monitored, as well as a wider review of the FEAT assessment process to ensure previous assessment results remain relevant.
- **Review and approval:**
  - Higher risk use cases would necessitate more stringent review and approval processes, which may be reflected in the number and levels of parties required for sign off.
- **Escalation process:**
  - Deviations from pre-agreed model performance or FEAT metric thresholds under higher risk use cases call for greater levels of escalation and intervention to address the matter.
- **Internal vs. external third party audit:**
  - Higher risk use cases may benefit from independent validation through third party audits.
  - For lower risk use cases, it may be adequate to conduct internal audits where in house expertise is available.

### 3.2.4 Integrating the FEAT Principles Assessment Methodology with Existing Risk Management Practices

Banks traditionally have well defined Model Risk Management (MRM) practices embedded into their processes for managing, monitoring, and auditing risks (as defined by the three lines of defence). However, these risk practices and governance setup focus on reducing the risk of material financial loss to the institution from material models, they do not incorporate elements of the FEAT Principles. Globally, regulatory agencies are evaluating the suitability of existing laws, regulations, and guidance for AIDA models. For example, guidance issued by the European Central Bank (ECB) Supervisory Board in 2018 recognises the key challenges with respect to AIDA system risk including bias and explainability.[7]

While traditional MRM approaches (such as outcome analysis), have accounted for issues in model interpretability and bias mitigation, the use of increasingly varied data coupled with black box models requires enhancing MRM reach across organisations and therefore across use cases with varying levels of materiality. Outcome assessment entails the evaluation of factors such as the overall model design/methodology, development/estimation processes, assumptions/limitations, and the final model specification. In the context of AIDA models, there needs to be an emphasis on the consideration of alternate methodologies and the value addition of AIDA approaches over more transparent methodologies.

The following illustrates how the recommendations and frameworks made above could be brought together to enhance traditional MRM components:

| MRM Components | Traditional view | Incorporating FEAT in MRM for AIDA systems |
|---|---|---|
| **Scope and applicability for MRM** | • Focused on risk management or regulatory/compliance use cases. | • Expand application to all use cases which use AIDA systems across the organisation (e.g., marketing, operations, etc.).<br>• Incorporate a risk-based approach (outlined in section 3.2.2) to incorporate the FEAT Principles in defining the scope and applicability of model approvals.<br>• Outline the FEAT implications and impact associated with high-risk AIDA models, and the enhanced review standards required (outlined in section 3.2.3).<br>• Implement changes to the approval and attestation process. |
| **Model inventory** | • Capture of key model information.<br>• Tracking of validation activities and the timeframe of model validity.<br>• Emphasis is placed on the process of tracking and documentation. | • Model information catalogue (including model usage, goals, owners, data controller/processor etc.,) to ensure traceability and transparency.<br>• Organisation-wide model identification based on consistent definitions, specifically for AIDA systems with black box interpretability requirements. |

| MRM Components | Traditional view | Incorporating FEAT in MRM for AIDA systems |
|---|---|---|
| **Model development, validation and implementation** | • Focus on data quality, conceptual soundness, outcome analysis and periodic monitoring.<br>• Documentation and tracking of activities across model lifecycle.<br>• Implementation is handled by IT teams. | • Adequate coverage of model purpose, usage and conceptual soundness.<br>• Ensuring appropriate and responsible usage of available data.<br>• Along with performance KPIs, provision model testing and monitoring for fairness and unintended outcomes.<br>• Incorporate FEAT Assessment Methodology checklist throughout the AIDA system development lifecycle.<br>• Adequate coverage and transparency considerations for third party (vendor) risks |
| **Risk tiering and assessment** | • Focus on model materiality and criticality considerations. | • Develop new lenses to define model risk tiering based on the enhanced risk associated with AIDA systems assessed against the FEAT Principles (outlined in section 3.2.2).<br>• Develop a risk specific application of the FEAT Principles development/validation frameworks. |
| **Governance standards** | • Enterprise wide view of model risk at board level.<br>• Business unit ownership for model risk.<br>• Clearly defined roles and responsibilities. | • Incorporate guiding FEAT Principles in the organisation's governance strategy.<br>• Ensure the governance framework is anchored to the organisation's core values, ethical guardrails, and accountability frameworks.<br>• Define measurable and trackable KPIs and metrics to ensure compliance to the organisation's overall values and principles.<br>• Define RACI for accountable parties, set processes for coordination and share effective communication standards across the teams involved (as outlined in section 3.2.1) |
| **People and technology development** | • Relevant product and technical skills at business unit level.<br>• Alignment of models with capabilities of the information system. | • Identify required skills/new roles aligned to the application of the FEAT Principles, and define training and resourcing needs across all lines of defence.<br>• Embed FEAT skills within teams to help escalate issues that need attention.<br>• Architect and deploy AIDA models, systems, and platforms that are trustworthy and fair by design.<br>• Establish technical methods and tools to assess AIDA systems before "go live" and monitor them afterwards (as outlines in section 3.2.1). |

*Table 3.1: Application of FEAT into MRM components*

# 04 Summary of FEAT Principles Assessment Methodology

This section summarises the deep dive methodology documents to give a high-level overview of the FEAT Principles Assessment Methodology to assess AIDA systems.

## 4.1 Fairness

As a concept, fairness is not novel to decision making and certainly not unique to Artificial Intelligence & Data Analytics driven decision making ("AIDA", which is defined by the Monetary Authority of Singapore as "technologies that assist or replace human decision-making"). In a general sense, fairness involves what is right, impartial and equal without favouritism or discrimination.

AIDA systems powered by high quality data, sophisticated algorithms and ever-increasing processing capabilities bring scale and competitive value to the data driven decision making processes. However, without careful design and control, these systems can bring new risks and unintended harms, perpetuating or reinforcing existing disadvantages in society or even introducing new ones. With so many decisions in financial services aided by AIDA systems, the fairness of these systems is becoming increasingly important.

More and more regulators worldwide are playing a critical role in guiding FSIs to prevent harms and deliver benefits. In Singapore, within the foundational FEAT Principles, the Principles that explicitly relate to fairness (see figure P1-4 in section 2.4) aim to ensure that AIDA driven decisions do not systematically disadvantage individuals or groups of individuals, without appropriate justification.

To help organisations apply the Principles, Veritas Phase 1 (published in January 2021) formulated the initial FEAT Fairness Principles Assessment Methodology. In Phase 1, use case examples of the "Fairness Assessment Methodology" focused on the banking sector. This document is the output of Veritas Phase 2 and it extends the Phase 1 Fairness Assessment Methodology with frameworks, practical guidance and examples, covering both banking and insurance. The Fairness Assessment Methodology provides a guide for FSIs to assess alignment of their AIDA systems' operations with the Fairness Principles.

The main extensions to the Fairness Assessment Methodology in Phase 2 are:

- Key concepts such as personal attributes, types of bias and mitigation methods, and fairness objectives and metrics are defined in more detail along with illustrative frameworks and guidance on how to identify and apply them to a use case.
- Parts A-E of the Fairness Assessment Methodology defined in Phase 1 are aligned to a typical AIDA system development lifecycle, enabling FSIs to develop fair AIDA applications by design.

- Detailed considerations and guidance are included for each of the 18 Fairness Assessment Methodology assessment questions from Phase 1 to help FSIs think through the important aspects to consider when answering the questions.
- The 18 Fairness Principles Assessment Methodology questions from Phase 1 are integrated into the overall FEAT checklist to enable a full FEAT assessment of the end-to-end AIDA system.
- Guidance is provided on scaling and embedding the Fairness Principles Assessment Methodology across an FSI, through
  - Defining the internal foundational set of practices and standards for AIDA fairness,
  - Including guidance on how to customise the depth of the fairness assessment required for a given AIDA system, and
  - Including guidance on how FSIs can integrate AIDA fairness into the fabric of their organisation.

The use case example in Phase 2 is predictive underwriting for life insurance. A simplified underwriting journey for existing customers of an insurer in Singapore and a detailed illustration of the use case can be found in the Veritas Phase 2 - Illustrative Case Studies document.

## 4.2 Ethics and Accountability

With the increased use of AIDA systems in the financial sector, the expectations for accountability and ethical behaviour from regulators, peers, consumers, and society as a whole have also increased. The material in the Ethics and Accountability Methodology document, and the accompanying workbook Operationalising Ethics and Accountability: Workbook (refer to appendix in the Ethics and Accountability Methodology document), can help to ensure consistent and context sensitive ethical decision making throughout organisations.

While it is typical for businesses to have a set of core values in place, and in some cases AI principles, there has been little progress in applying these values to decision making. However, FSIs have a universally strong foundation of governance in place to manage material and systemic risk, and are therefore better positioned to apply values and principles to decision making than companies in other sectors. This foundation can be leveraged with the framework contained in this document to govern ethical decision making and generate both internal and external accountability.

The Operationalising Ethics and Accountability framework, provides a taxonomy of terms to organise the various concepts that are important to ethical decision making and to move the organisation from abstract notions to concrete and measurable commitments to action. These terms include the values and principles that organisations are most familiar with as well as additional controls such as normative concepts, such as fairness, that serve to clarify, shape, and ground commitments and specifications, which are the specific policies and metrics, respectively, that enable organisations to be held accountable.

With this framework and the accompanying workbook, FSIs will be able to engage in measurable ethical practices that are consistent with organisational concepts of ethics and accountability. The intended audience could include a wide set of stakeholders, depending on the work being undertaken. If the organisation's core values need to be refreshed, then the office of the CEO will be involved; if it is used to specify the local context for fairness, then data scientists, business process owners, and/or AI management might need to be involved. The nature of the work and how it fits in existing risk frameworks and governance processes are all determinants for the intended audience and these will likely vary among FSIs.

## 4.3 Transparency

With many decisions in the Financial Services Industry (FSI) now being shaped or even made by algorithms, there is a greater demand for transparency around crucial questions (e.g., where are algorithms in use? what data do they use? how are they making decisions? what impact do they have? etc.)

Regulators around the world are encouraging FSIs to increase transparency around AIDA decisions for people who are impacted by such decisions. This includes three of the 14 principles (see section 2.4 for the list of Principles) under the Monetary Authority of Singapore's FEAT guidelines released in 2018.

FSIs can choose to view transparency only from a regulatory compliance perspective. However, to do so would be to waste an important opportunity. In an increasingly digital world, transparency is an unmet customer need, and providing greater transparency could help FSIs build stronger customer relationships and create competitive advantage. In addition, creating greater internal transparency into algorithms can help overcome the trust deficit that AIDA systems may suffer within the FSI and/or with regulators.

FSIs' approach to AIDA transparency must address two aspects:

- **External.** What level and form of transparency should FSIs provide to customers/prospects and other external stakeholders in various situations?
- **Internal.** How should FSIs go about building reliable explanations to feed into (and provide verifiable backup for) the information shared externally and to be compliant with relevant internal policies and procedures and regulatory requirements?

With respect to external transparency, a variety of regulatory considerations apply, such as existing ones around treating customers fairly and respecting an individual's personal data rights, and new ones specific to AIDA systems. As a result, it is difficult to prescribe the level of transparency needed for different AIDA use cases.

However, it is possible to provide a checklist of the key considerations for an FSI in arriving at such a decision. The materiality of the AIDA driven decision to the impacted customer (e.g., whether it results in denial of access to financial service) must be weighed against considerations such as the prevention of fraud or gaming by unscrupulous applicants or the FSI's legal obligations such as financial crime prevention/detection.

There are also practical considerations to be kept in mind in the operationalisation of transparency. For example, human beings are limited in their ability to interpret complex explanations for algorithmic decisions. The reasons behind a decision may be helpful in educating them, but sometimes it may be more valuable to highlight the actions that they can take to change the outcome.

With regard to internal transparency, the past few years have seen significant activity around designing explanation methods for AIDA systems, particularly with respect to ML models. FSIs should review the existing landscape of methods and consider implementing internal standards for approved explanation methods.

More broadly, FSIs should take steps to implement transparency requirements in a scalable way, rather than on a case-by-case basis. Transparency methodology document makes five recommendations in this regard: embedding AIDA transparency considerations into existing risk frameworks; investing in proactive communication; training customers, staff, management and risk/ control functions; investing in tools for AIDA transparency; and integrating transparency considerations throughout the AIDA lifecycle.

Finally, it should be recognised that the field of algorithmic transparency is relatively immature. There are multiple areas in which there is room for future work around regulatory guidance, industry practices and even academic research (e.g., the alignment of transparency considerations between AIDA and non-AIDA systems, or the treatment of third party data and models with respect to transparency requirements).



# 05 Global Regulatory Landscape on Regulating Responsible AI

## 5.1 Global Regulatory Landscape

Various regulatory agencies across the world are formulating frameworks to address the unique opportunities and challenges presented by AIDA systems. Recent regulatory developments include:



### Singapore

- The Model AI Governance Framework.[8]
- Implementation and Self-Assessment Guide for Organisations.[9]
- Compendium of Use Cases.[10]



### Hong Kong

- High Level Principles on Artificial Intelligence.[11]
- Consumer Protection in Respect of Use of Big Data Analytics and AI by Authorised Institutions.[12]



### United States

- Five regulators issued a request for information and comment on the use of AI and ML by FSIs.[13]
- US National Association of Insurance Commissioners AI Guiding Principles.[14]



### United Kingdom

- The Bank of England Future of Finance Report.[15]
- The opening of the Artificial Intelligence Public-Private Forum (AIPPF) to facilitate a dialogue with the private sector.[16]



### The European Union

- European Commission has proposed the first regulatory framework for AI: the Proposal for a Regulation on a European approach for Artificial Intelligence[17]

## 5.2 Key Regulatory Trends

The overall trend in the regulatory landscape has been a focus on clarifying and revising current laws and standards so they can be more clearly applied to circumstances involving AIDA systems, rather than on establishing new regulatory bodies and laws.[18]

Regulators in Asia have issued non-binding guidelines and principles on the adoption of AI, endorsing a principles based, technology neutral approach to the responsible use of AI. This regulatory approach can be seen most clearly in Singapore and Hong Kong, which is similar to the approach taken outside Asia by jurisdictions including the UK.

## 5.3 Assessing the Regulatory Implications of AIDA Projects in Financial Services

Different legal and regulatory requirements will apply to the different contexts in which FSIs are using AIDA, so there is no one-size-fits all approach. Table 5.1 summarises learnings in the form of key considerations for FSIs when assessing the regulatory implications of their development and deployment of AIDA solutions.

| | |
|---|---|
| 1 | Adopt a materiality based approach to assess regulatory considerations when AIDA is being used to help focus efforts on more sensitive use cases. |
| 2 | Leverage existing internal risk management and governance frameworks when assessing regulatory implications of using AIDA in specific use cases. |
| 3 | Collaborate with technology partners to promote transparency and accountability when deploying AIDA systems. |
| 4 | Bring a diverse set of perspectives to discussions on regulatory issues relating to Responsible AI to ensure ethics principles are appropriately applied, and that potential concerns are identified and dealt with early. |
| 5 | Share use cases and examples where possible to strengthen understanding across the industry of what regulatory issues may apply when using AIDA |

*Table 5.1: Key considerations for FSIs when assessing regulatory implications of AI*

## 5.4 Adopting a Materiality Based Approach to Assessing Regulatory Considerations

The materiality based approach has been covered in detail in section 3.2.2 of this document. The risk based approach has been endorsed by MAS in the publication of its FEAT Principles,[19] where firms are encouraged to calibrate actions and requirements under their internal governance framework based on the materiality of the AIDA driven decisions. Some of the suggested considerations to determine materiality under these principles include the extent to which the AIDA system is used in decision making, the extent of automation of the process, and the severity and probability of impact on different stakeholders. The materiality based approach can also be seen in the European Commission's Proposal for a Regulation on a European approach for Artificial Intelligence.

## 5.5 Leverage Existing Risk Management and Governance Frameworks

Existing risk management and governance frameworks are a useful starting point for FSIs to consider when implementing measures for the responsible use of AI. FSIs already operate in a highly regulated industry. Most jurisdictions already have robust regulatory requirements in place covering outsourced technology use, risk management, and governance. These regulations are almost always technology neutral, meaning that they apply regardless of whether AI is used or not.

A key task for assessing the regulatory implications when implementing AIDA solutions would therefore be to identify the regulatory issues in a specific use case. Table 5.2 gives a hypothetical example of working through some of the relevant regulatory questions involved in an AI context, noting relevant Singapore regulatory requirements that may apply.

| Phase | Some regulatory considerations | Jurisdictional examples |
|---|---|---|
| **Phase 1: input of data into AI technology** | • Data protection laws.[20]<br>• Confidentiality laws.<br>• Guidelines on outsourcing.[21] | While Singapore's Personal Data Protection Act (No. 26 of 2012)[22] does not have a specific provision relating to the use of AI, AIDA applications' reliance on internal and/or external data means that sector agnostic compliance requirements such as data protection regulation remains relevant. FSIs must therefore comply with the PDPA's framework on the collection, use and disclosure of personal data by private sector organisations in Singapore to protect the personal data of individuals, support public trust in the digital economy and enable innovation in the data space. |

| Phase | Some regulatory considerations | Jurisdictional examples |
|---|---|---|
| **Phase 2: processing of data by AI** | • Technology laws.<br>• Cybersecurity Laws.<br>• Risk management practices. | MAS has published several risk management practice guidelines, such as the Guidelines on Risk Management Practices for Insurance Business - Core Activities,[23] the Guidelines on Risk Management Practices - Technology Risk, and the Guidelines on Risk Management Practices – Board and Senior Management.[24] MAS generally requires all FSIs in Singapore to implement and maintain risk management frameworks, and to regularly update and develop such frameworks when new risks are identified. With the increasing adoption of AI solutions by FSIs, it is important for FSIs to evaluate the risks that the use of AI solutions may pose to their businesses. Thereafter, FSIs should update their existing risk management frameworks to include additional measures to address such new risks. |
| **Phase 3: output generated by the AI use case** | • Consumer protection laws.<br>• Anti-discrimination laws. | Singapore's Guidelines on the Provision of Digital Advisory Services[25] applies to FSIs offering or seeking to offer digital advisory services in Singapore. Digital advisors must put in place adequate governance and supervisory arrangements to effectively mitigate against any additional risks created by digital advice (e.g., that a fault or bias in an algorithm could adversely affect a large number or all of the digital advisors' clients who relied on the recommendations generated by the algorithms). |

*Table 5.2: Hypothetical example of regulatory considerations in Singapore in an AI use case*

## 5.6 Collaborate with Technology Partners

A growing area of interest is in the complementary roles of FSIs deploying AIDA systems, and Technology firms providing AI solutions that FSIs might either customise or deploy "off-the-shelf". Technology firms can share technical and non-technical information with their customers about the capabilities and limitations of AI software to assist the responsible deployment of AI products, and promote transparency and accountability.

Anecdotally, regulators are encouraging collaboration between FSIs and technology partners to improve understanding of how existing internal risk management and governance frameworks can be leveraged to implement and demonstrate good practices of Responsible AI. There are also several formal relationships emerging between regulators, technology companies and FSIs. For example, like the multistakeholder collaboration of the Veritas Consortium in Singapore, the European Commissions' Ethics Guideline for Trustworthy AI[26] was prepared by members of prominent technology/financial companies such as AXA, Microsoft and Bosch, as well as universities, and government advisers.

## 5.7 Ensure a Diversity of Perspectives

A diverse set of perspectives is of upmost importance in discussing regulatory issues and helps to ensure that Responsible AI principles are appropriately applied, and that potential concerns are identified and dealt with early.

Diversity can come in various forms. For one, collaboration with a range of stakeholders (e.g., governments, regulators, industry associations, technology developers, etc.) can help promote transparency and accountability and ensures that principles can be translated into practical and implementable actions. The Veritas consortium, as well as the EU's Ethics Guideline for Trustworthy AI[27] are examples of this approach (see 5.6 above).

Diversity can also be enhanced through engagement with different teams within an organisation itself, such as the teams responsible for developing the AI solutions, as well as the legal, risk and compliance teams. Diversity should also be reflected within each team in terms of gender, ethnic groups, disability, race, etc., to ensure a diverse set of views are contributing to any decisions around the implementation of AI.

## 5.8 Share Use Cases with the Industry

Given that it is largely the context in which an AIDA system is being used that triggers regulatory implications (i.e., not the use of system alone) sharing use cases and examples can strengthen understanding across the industry of what regulatory issues may apply. Initiatives like Veritas are valuable in creating opportunities to share use cases in this way.

The importance of use case sharing was in evidence in the multistakeholder collaboration between Microsoft, Standard Chartered, Deutsche Bank, Visa and Linklaters[28] to test the application of the FEAT Principles. These industry participants worked collectively to assess the FEAT considerations involved in three different financial services uses cases. The project partners found specific Principles have different implications in different contexts, and some principles will be more, or less, relevant depending on the use case.

## 5.9 The Way Forward for Veritas

With the rapid increase in deployment of AIDA systems by FSIs in Asia, a key reason for implementing guidance like MAS' FEAT Principles is that such regulator guidance provides a baseline set of Responsible AI criteria, which helps FSIs ask the right questions when deploying AI solutions. Guidelines on the responsible use of AI should acknowledge the importance of context in translating high level principles into practice. An important learning from discussions in Veritas and beyond is that each Principle will have different implications depending on the context, and a materiality assessment will be important to determine which Principle is relevant and how risks can be mitigated.

For a little more detail on the global regulatory landscape, please refer to Appendix C.

# 06 Acknowledgements

| Role | Name | Organisation |
|---|---|---|
| Project Sponsor | Sopnendu Mohanty | Monetary Authority of Singapore |
| Project Director | Li XuChun | Monetary Authority of Singapore |
| Project Lead | Zhang Qiang | Monetary Authority of Singapore |
| Project Manager | Bhavna Rawlley | Accenture |
| Data Scientist | Rupini Pandian | Accenture |
| Data Scientist | Shen Kai | Accenture |
| Subject Matter Expert | Marcus Bartley Johns | Microsoft |
| Subject Matter Expert | Dennae Smith | Microsoft |
| Subject Matter Expert | Nick Lewins | Microsoft |
| Subject Matter Expert | Dave Dadoun | Microsoft |

We would also like to express our appreciation to James Hwa Jaan Gan (Accenture) and Joon Seong Lee (Accenture) who have supported and contributed to this project.

# 07 Bibliography


[1] Monetary Authority of Singapore (2021). Veritas Phase 1 document. Retrieved from https://www.mas.gov.sg/-/media/MAS/News/Media-Releases/2021/Veritas-Document-1-FEAT-Fairness-Principles-Assessment-Methodology.pdf

[2] Monetary Authority of Singapore (2021). Veritas Phase 1 document. Retrieved from https://www.mas.gov.sg/-/media/MAS/News/Media-Releases/2021/Veritas-Document-2-FEAT-Fairness-Principles-Assessment-Case-Studies.pdf

[3] Ferrante, KH., D'Orazio, V., & Sicular, S. (2021). Financial Services CIOs Must Focus AI Investments on 'Responsible AI' in 2021, Gartner. Retrieved from https://www.gartner.com/en/documents/3995169/financial-services-cios-must-focus-ai-investments-on-res

[4] Awalegaonkar, K., Berkey, R., Douglass, G., & Reilly, A. (2018). "Critical mass: Managing AI's unstoppable progress.", Accenture. Retrieved from https://www.accenture.com/ch-en/insights/artificial-intelligence/ai-investments

[5] Monetary Authority of Singapore (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/publications/monographs-or-information-paper/2018/FEAT

[6] Monetary Authority of Singapore (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/~/media/MAS/News%20and%20Publications/Monographs%20and%20 Information%20Papers/FEAT%20Principles%20Final.pdf

[7] European Central Bank (2018). The digitalisation of banking – supervisory implications. Retrieved from https://www.bankingsupervision.europa.eu/press/speeches/date/2018/html/ssm.sp180606.en.html

[8] Personal Data Protection Commission (2020). Model Artificial Intelligence Governance Framework. Retrieved from https://www.pdpc.gov.sg/-/media/Files/PDPC/PDF-Files/Resource-for-Organisation/AI/SGModelAIGovFramework2.pdf

[9] Personal Data Protection Commission (2020). Companion to the Model AI Governance Framework. Retrieved from https://www.pdpc.gov.sg/-/media/Files/PDPC/PDF-Files/Resource-for-Organisation/AI/SGIsago.pdf

[10] Personal Data Protection Commission (2020). Compendium of Use Cases: Practical illustrations of the model AI governance framework. Retrieved from https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgaigovusecases.pdf

[11] Hong Kong Monetary Authority (2019). Retrieved from https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2019/20191101e1.pdf

[12] Hong Kong Monetary Authority (2019). Retrieved from https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2019/20191105e1.pdf

[13] Federal Register, The Daily Journal of United States Government (2021). Request for Information and Comment on Financial Institutions' Use of Artificial Intelligence, Including Machine Learning. Retrieved from https://www.federalregister.gov/documents/2021/03/31/2021-06607/request-for-information-and-comment-on-financial-institutions-use-of-artificial-intelligence

[14] The National Association of Insurance Commissioner (2020). Retrieved from https://content.naic.org/article/news_release_naic_unanimously_adopts_artificial_intelligence_guiding_principles.htm

[15] Bank of England (2019). Review on the outlook for the UK financial system: what it means for the Bank of England. Retrieved from https://www.bankofengland.co.uk/-/media/boe/files/report/2019/future-of-finance-report.pdf?la=en&hash=59CEFAEF01C71AA551E7182262E933A699E952FC

[16] Bank of England (2020). Fintech AI Public – Private Forum. Retrieved from https://www.bankofengland.co.uk/events/2020/october/fintech-ai-public-private-forum

[17] European Commission (2021). Laying down harmonized rules on Artificial Intelligence (Artificial Intelligence Act) and amending certain union legislative acts. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1623335154975&uri=CELEX%3A52021PC0206

[18] Evans, CL. (2020). Artificial Intelligence and Public Standards. Retrieved from https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/868284/Web_Version_AI_and_Public_Standards.PDF

[19] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/~/media/MAS/News%20and%20Publications/Monographs%20and%20Information%20Papers/FEAT%20Principles%20Final.pdf

[20] Prenio, J., & Yong, J., (2021). Humans keeping AI in check – emerging regulatory expectations in the financial sector. Retrieved from https://www.bis.org/fsi/publ/insights35.pdf

[21] Prenio, J., & Yong, J., (2021). Humans keeping AI in check – emerging regulatory expectations in the financial sector. Retrieved from https://www.bis.org/fsi/publ/insights35.pdf

[22] Singapore Statutes Online (2012). Personal Data Protection Act 2012. Retrieved from https://sso.agc.gov.sg/Act/PDPA2012

[23] Monetary Authority of Singapore (2013). Guidelines on Risk Management Practices for Insurance Business – Core Activities. Retrieved from https://www.mas.gov.sg/-/media/MAS/Regulations-and-Financial-Stability/Regulatory-and-Supervisory-Framework/Risk-Management/Risk-Management-Guidelines_Insurance-Core-Activities.pdf

[24] Monetary Authority of Singapore (2013). Guidelines on Risk Management Practices – Board and Senior Management. Retrieved from https://www.mas.gov.sg/-/media/MAS/Regulations-and-Financial-Stability/Regulatory-and-Supervisory-Framework/Risk-Management/Board-and-Senior-Mgmt_1-Apr-2013.pdf

[25] Monetary Authority of Singapore (2018). Guidelines on Provision of Digital Advisory Services. Retrieved from https://www.mas.gov.sg/-/media/MAS/Regulations-and-Financial-Stability/Regulations-Guidance-and-Licensing/Securities-Futures-and-Fund-Management/Guidelines-on-Provision-of-Digital-Advisory-Services--CMGG02.pdf

[26] European Commission (2019). Ethics guidelines for trustworthy AI. Retrieved from https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[27] European Commission (2019). Ethics guidelines for trustworthy AI. Retrieved from https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[28] Microsoft's collaboration with Standard Chartered, Deutsche Bank, Visa and Linklaters (2019). From Principles to Practice – Use Cases for Implementing Responsible AI in Financial Services. Retrieved from https://www.microsoft.com/cms/api/am/binary/RE487kh

[29] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[30] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[31] Hancock, J., Huber, P., & Koch, P., (1999). The economics of insurance: how insurers create value for shareholders. Retrieved from https://www.worldcat.org/title/economics-of-insurance-how-insurers-create-value-for-shareholders/oclc/83511751

[32] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[33] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[34] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[35] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[36] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[37] Minty, D., (2020). Behavioural fairness is a serious risk to the future of insurance. Retrieved from https://ethicsandinsurance.info/2020/04/29/behavioural-fairness/

[38] European Commission (2019). Ethics Guidelines for Trustworthy AI. Retrieved from https://www.europarl.europa.eu/cmsdata/196377/AI%20HLEG_Ethics%20Guidelines%20for%20Trustworthy%20AI.pdf

[39] Bias. Retrieved from https://en.wikipedia.org/wiki/Bias

[40] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[41] Jenkins, OB. (2012). The Meaning of "Fairness". Retrieved from http://www.orvillejenkins.com/faithlife/fairnessfl.html

[42] Mulligan, D., Kroll, J., Kohli, N. & Wong, R. (2019). This thing called fairness: Disciplinary confusion realizing a value in technology. ACM Human-Computer Interaction, 3, 119. Retrieved from https://doi.org/10.1145/3359221

[43] Smith, G., Kohli, N., & Rustagi, I. (2020). Mitigating Bias in Artificial Intelligence: An Equity Fluent Leadership Playbook. Retrieved from https://haas.berkeley.edu/wp-content/uploads/What-is-fairness_-EGAL2.pdf

[44] Monetary Authority of Singapore. (2018). Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of AI and Data Analytics in Singapore's Financial Sector. Retrieved from https://www.mas.gov.sg/-/media/MAS/News-and-Publications/Monographs-and-Information-Papers/FEAT-Principles-Updated-7-Feb-19.pdf

[45] Atlantic Council (2021). Getting from commitment to content in AI and data ethics: Justice and explainability. Retrieved from https://www.atlanticcouncil.org/in-depth-research-reports/report/specifying-normative-content/

[46] Market Business News (2021). Financial Glossary. Retrieved from https://marketbusinessnews.com/financial-glossary/objective/

[47] Personal Data Protection Commission (2020). Model Artificial Intelligence Governance Framework. Retrieved from https://www.pdpc.gov.sg/-/media/Files/PDPC/PDF-Files/Resource-for-Organisation/AI/SGModelAIGovFramework2.pdf

[48] Personal Data Protection Commission (2020). Companion to the Model AI Governance Framework. Retrieved from https://www.pdpc.gov.sg/-/media/Files/PDPC/PDF-Files/Resource-for-Organisation/AI/SGIsago.pdf

[49] Personal Data Protection Commission (2020). Compendium of Use Cases: Practical illustrations of the model AI governance framework. Retrieved from https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgaigovusecases.pdf

[50] Hong Kong Academy of Finance (2020). Artificial Intelligence in Banking: The changing landscape in compliance and supervision. Retrieved from https://www.aof.org.hk/docs/default-source/hkimr/applied-research-report/airep.pdf

[51] Steenis, Hv. (2019). Future of Finance. Retrieved from https://www bankofengland.co.uk/-/media/boe/files/report/2019/future-of-finance-report.pdf?la=en&hash=59CEFAEF01C71AA551E7182262E933A699E952FC

[52] Federal Register, The Daily Journal of United States Government (2021). Request for Information and Comment on Financial Institutions' Use of Artificial Intelligence, Including Machine Learning. Retrieved from https://www.federalregister.gov/documents/2021/03/31/2021-06607/request-for-information-and-comment-on-financial-institutions-use-of-artificial-intelligence

[53] The National Association of Insurance Commissioner (2020). Retrieved from https://content.naic.org/article/news_release_naic_unanimously_adopts_artificial_intelligence_guiding_principles.htm

[54] European Commission (2021). Laying down harmonised rules on Artificial Intelligence (Artificial Intelligence Act) and amending certain union legislative acts. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1623335154975&uri=CELEX%3A52021PC0206

[55] Evans, CL. (2020). Artificial Intelligence and Public Standards. Retrieved from https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/868284/Web_Version_AI_and_Public_Standards.PDF

[56] Hong Kong Monetary Authority (2019). Retrieved from https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2019/20191101e1.pdf

[57] Steenis, Hv. (2019). Future of Finance. Retrieved from https://www.bankofengland.co.uk/-/media/boe/files/report/2019/future-of-finance-report.pdf?la=en&hash=59CEFAEF01C71AA551E7182262E933A699E952FC

[58] Crampton, N. (2021). The building blocks of Microsoft's responsible AI program. Retrieved from https://blogs.microsoft.com/on-the-issues/2021/01/19/microsoft-responsible-ai-program/

[59] European Commission (2021). Laying down harmonised rules on Artificial Intelligence (Artificial Intelligence Act) and amending certain union legislative acts. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1623335154975&uri=CELEX%3A52021PC0206

[60] Microsoft's collaboration with Standard Chartered, Deutsche Bank, Visa and Linklaters (2019). From Principles to Practice – Use Cases for Implementing Responsible AI in Financial Services. Retrieved from https://www.microsoft.com/cms/api/am/binary/RE487kh

[61] Prenio, J., & Yong, J., (2021). Humans keeping AI in check – emerging regulatory expectations in the financial sector. Retrieved from https://www.bis.org/fsi/publ/insights35.pdf

[62] Singapore Statutes Online (2012). Personal Data Protection Act 2012. Retrieved from https://sso.agc.gov.sg/Act/PDPA2012

[63] Monetary Authority of Singapore (2013). Guidelines on Risk Management Practices for Insurance Business – Core Activities. Retrieved from https://www.mas.gov.sg/-/media/MAS/Regulations-and-Financial-Stability/Regulatory-and-Supervisory-Framework/Risk-Management/Risk-Management-Guidelines_Insurance-Core-Activities.pdf

[64] Monetary Authority of Singapore (2013). Guidelines on Risk Management Practices – Board and Senior Management. Retrieved from https://www.mas.gov.sg/-/media/MAS/Regulations-and-Financial-Stability/Regulatory-and-Supervisory-Framework/Risk-Management/Board-and-Senior-Mgmt_1-Apr-2013.pdf

[65] Monetary Authority of Singapore (2018). Guidelines on Provision of Digital Advisory Services. Retrieved from https://www.mas.gov.sg/-/media/MAS/Regulations-and-Financial-Stability/Regulations-Guidance-and-Licensing/Securities-Futures-and-Fund-Management/Guidelines-on-Provision-of-Digital-Advisory-Services--CMGG02.pdf

[66] European Commission (2019). Ethics guidelines for trustworthy AI. Retrieved from https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[67] European Commission (2019). Ethics guidelines for trustworthy AI. Retrieved from https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[68] Microsoft's collaboration with Standard Chartered, Deutsche Bank, Visa and Linklaters (2019). From Principles to Practice – Use Cases for Implementing Responsible AI in Financial Services. Retrieved from https://www.microsoft.com/cms/api/am/binary/RE487kh

[69] Andrews, K. (2019). Businesses ready to test AI ethics principles. https://www.minister.industry.gov.au/ministers/karenandrews/media-releases/businesses-ready-test-ai-ethics-principles

# 08 Appendix A

## FEAT Principles Assessment Methodology Checklist

### 1 Checklist for Assessing Adherence to FEAT Throughout the AIDA System Development Lifecycle

As highlighted previously, this checklist is optional and is only relevant if an FSI chooses to take steps to adhere to FEAT. It should be recognised that making AIDA systems fair, ethical, accountable and transparent is not a trivial task. Therefore, a certain number of checklist items are inevitable: comparable checklists, like the one in the European Commission's Ethics Guidelines for Trustworthy AI, have more than 30 items. Trying to overly simplify or rationalise the checklist will be counterproductive and could produce false sense of comfort.

However, the sub-questions (provided in the individual methodology documents) underneath these top-level checklist questions should be seen as considerations to help answer those questions. FSIs may wish to track adherence through the top level checklist items rather than the more detailed considerations below. If FSIs decide to follow FEAT principles, the evidence to this checklist should be available on demand if required.

### 1.1 Diagram to Align the Questions with the Respective Stage

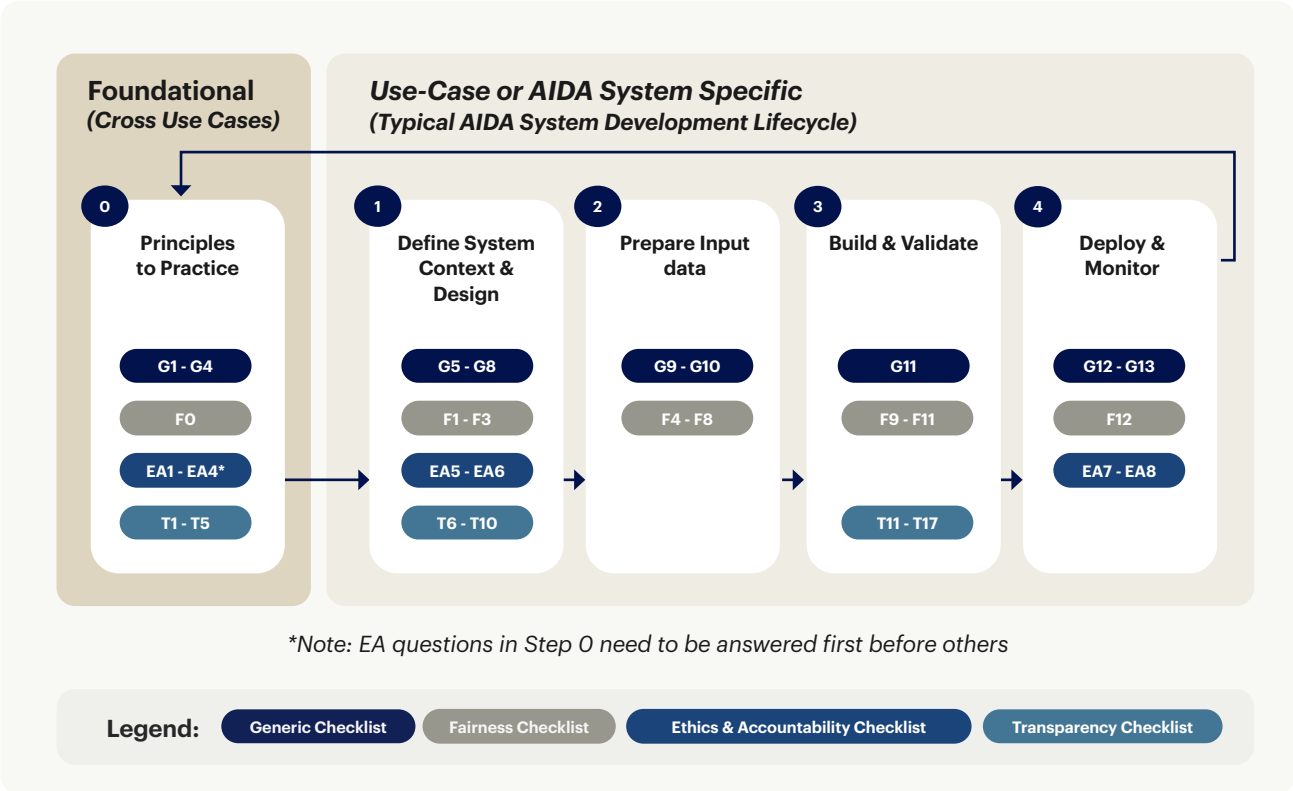



*Figure 1.1 FEAT assessment checklist mapped against the AIDA system development lifecycle*

## 1.2 Detailed Questions

### 1.2.1 Foundational Setup

This part of the checklist creates a foundation for ensuring the right implementation of FEAT Principles across AIDA use cases. This part only needs to be answered once initially. Ethics & Accountability defines how an organisation's values are translated to FEAT and is a starting point for organisation's FEAT journey. Once the initial Ethics & Accountability checklist have been completed, the organisation can proceed to define and assess Fairness, Transparency and overall AIDA systems.

| S/N | Question | Y/N |
|---|---|---|
| **Step 0 – Principles to Practice** | | |
| **G1** | Is AIDA defined in the FSI? | |
| **G2** | Is there a framework in place to define roles and responsibilities for AIDA projects/use cases? | |
| **G3** | Is there a materiality framework in place? If so, do you have a clear definition to determine the different level of materiality?<br>*Refer to the FEAT Principles Assessment Methodology section 3.2.2 for parameters that could be used to determine risk level* | |
| **G4** | Does the FSI have an up-to-date inventory that tracks all material AIDA use cases in use? | |
| **EA1** | Are organisational values defined and described? | |
| **EA2** | Are organisational, or group-specific, principles for the ethical use of AI defined and described? | |
| **EA3** | Are context-relevant core concepts identified and described? | |
| **EA4** | Is the workforce trained on values-based decision making or is there a pilot planned? | |
| **T1** | Has the FSI defined the factors it will use to determine whether external (customer facing) transparency is essential for a particular AIDA use case? | |
| **T2** | *(Where an FSI has chosen to provide external transparency)* At each stage of the FSI's customer lifecycle, has the FSI determined what proactive or reactive communication may be needed, and developed standard templates/interfaces for the same? | |
| **T3** | Has the FSI defined the factors it will use to determine the extent of, and audience for, internal transparency for individual AIDA use cases? | |
| **T4** | Has the FSI defined an acceptable set of AIDA ML explanation method(s) for use within the FSI?<br>*This may vary by use case type and/or materiality.* | |
| **T5** | Has the FSI set minimum accuracy standards for such explanation methods?<br>*This may vary by use case type and/or materiality.* | |
| **F0** | Has the FSI defined standard, robust process for (a) identifying at risk groups? (b) identifying personal attributes and potential proxies?<br>(c) identifying explicit fairness objectives and associated measures and thresholds? and (d) algorithmic methods to identify and reduce unfairness? | |

## 1.2.2 FEAT Principles Assessment for Each AIDA System in Place

This part of the checklist is use case specific.

| S/N | Question | Y/N |
|---|---|---|
| **Step 1 - Defining System Context and Design** | | |
| **G5** | Have you documented the commercial objectives of the system and the quantitative measures to meet those commercial objectives? Is it documented how is AIDA used to achieve them? | |
| **G6** | Are there scope boundaries of the proposed system? For example, specific geographies/ segments? | |
| **G7** | Is AIDA use justified for this particular objective/use case (based on limitations of alternatives, proof of AIDA being used successfully elsewhere, availability of adequate data to train the AIDA system, valid legal basis, etc.)? | |
| **G8** | Has the materiality of the use case been determined using the framework defined in Step 0? | |
| **EA5** | Is there a statement of relevant values, core concepts, principles, commitments, and specifications, along with a written description, for each use case? *Part of this should include identifying commitment owners and the stakeholders with interest in holding the commitment owners accountable.* | |
| **EA6** | Have relative priorities, which may or may not be tied to materiality, been recorded for each commitment? | |
| **F1** | Have you identified and documented who are the individuals and groups that are considered to be at risk of being systematically disadvantaged by the system? | |
| **F2** | Have you identified and documented the potential harms and benefits created by the system's operation that are relevant to the risk of systematically disadvantaging the individuals and groups in F1? | |
| **F3** | Have you identified and documented the fairness objectives of the system and associated fairness metrics, with respect to the individuals and groups in F1 and the harms and benefits in F2? | |
| **T6** | Has the AIDA use case team determined whether there is a need for external (customer facing) transparency? *Apply standards from T1 to help answer the question.* | |
| **T7** | If yes, has the team identified the proactive and reactive communication needed at each stage of the customer lifecycle, and the form of such customer facing communication?<br>*Apply standards from T2 to help answer the question.* | |
| **T8** | Has the team determined the level of internal transparency needed, and the audiences for the same? *Apply standards from T3 to help answer the question.* | |
| **T9** | Has the team selected a suitable explanation method for this specific use case from the approved list in T4? | |
| **T10** | Has the team ascertained that the chosen explanation method/implementation meets the minimum accuracy requirement for this specific use case (based on T5)? | |

| S/N | Question | Y/N |
|---|---|---|
| **Step 2 - Prepare and Input Data** | | |
| **G9** | Have you documented the data used by the system including types of data available, the provenance (source), attributes, dictionary (meaning), vintage (time period), data (quality) profile, representativeness and informed consent? For any personal data used by the system, has a DPIA been conducted? | |
| **G10** | Have you documented the data pre-processing and engineering done by the system? | |
| F4 | Have you documented key errors, biases or properties present in the data used by the system that may impact the system's fairness? | |
| F5 | Have you documented how are these impacts being mitigated? | |
| F6 | Have you assessed and documented the quantitative estimates of the system's performance against its fairness objectives and the uncertainties in those estimates, assessed over the individuals and groups in F1 and the potential harms and benefits in F2? | |
| F7 | Have you assessed and documented the achievable trade-offs between the system's fairness objectives and its other objectives? | |
| F8 | Have you justified and documented why the fairness outcomes observed in the system are preferable are to these alternative trade-offs? | |
| **Step 3 - Build and Validate** | | |
| **G11** | Is the composition of the AIDA system defined? Is it clear how each component of the system AIDA models, as well as business rules and human judgement if relevant, are used to achieve its commercial objectives? Are the performance estimates and the uncertainties of those estimates documented? | |
| F9 | Have you assessed and documented the quantitative estimates of the system's performance against its fairness objectives and the uncertainties in those estimates, assessed over the individuals and groups in F1 and the potential harms and benefits in F2? | |
| F10 | Have you assessed and documented the achievable trade-offs between the system's fairness objectives and its other objectives? | |
| F11 | Have you justified and documented why the fairness outcomes observed in the system are preferable are to these alternative trade-offs? | |
| **T11** | Have internal transparency dashboards/reports been implemented in line with the requirements agreed in T8-T10? | |
| **T12** | Have relevant first and second line control teams – including model validation where relevant – reviewed and approved these outputs (e.g., local and global explanations, conceptual soundness)? | |
| **T13** | Where explanations have not met first/second line expectations, have appropriate mitigation actions been taken (e.g., switching to a simpler model despite a reduction in predictive accuracy, dropping difficult to explain features, introducing more human oversight)? | |
| **T14** | In line with the external transparency requirements agreed in T6-T7, has appropriate system functionality been developed and tested as part of the AIDA system's implementation plan? | |

| S/N | Question | Y/N |
|---|---|---|
| T15 | Have operational processes such as customer service and complaint handling been modified appropriately to incorporate AIDA customer transparency? Have relevant staff been provided appropriate training to address customer queries? | |
| T16 | Have customer/website Terms and Conditions been appropriately updated? | |
| T17 | Does the AIDA system implementation support the agreed internal and external explanations even after "go live" (i.e., not just as a one off before approval but throughout the lifetime of the AIDA system)? | |
| **Step 4 – Deploy and Monitor** | | |
| G12 | Is the system's monitoring and review regime designed to detect abnormal operation (e.g., significant model and/or data drift)? | |
| G13 | Is there fallback and/or mitigation plans in place in case of triggers from the system's monitoring and review regime? | |
| EA7 | Are recourse mechanisms available to data subjects being used? Is there an explanation if the use of recourse is too high or too low? | |
| EA8 | Are controls in place to revisit and recalibrate the commitments and their specifications to ensure they're properly incentivising the right decisions and accurately measuring progress toward the commitments? | |
| F12 | Does the system's monitoring and review regime ensure that the system's impacts are aligned with its business and fairness objectives? | |

# 09 Appendix B

## Glossary

| | |
|---|---|
| **Accountability** | The state of being responsible for a particular set of outcomes, such as behaviours, actions, products, services, or decisions. In combination with recourse and redress, it is a central component of any system of governance (see also: "recourse" and "redress"). |
| **Internal accountability**[29] | The ability of an organisation's governance system to hold an individual or group within the organisation responsible for unwanted outcomes. |
| **External accountability**[30] | The ability of governments, regulatory bodies, individuals, and other agencies to hold an organisation responsible for the outcomes of its operations. |
| **Accuracy** | A metric to evaluate the quality of a model. Often calculated as a percentage of predictions that are correct or incorrect, out of all cases considered. Can be used to indicate the effectiveness of a model compared to the status quo (see also: "model"). |
| **Actuarial fairness** | An example of quantitative fairness. Every insurance contract should contribute its full production cost (i.e., the cost directly attributable, like expected claims), and a fair (i.e., generally accepted and pre-defined) allocation of common cost, like allocated capital cost.[31] Actuarial fairness has been the guiding principle of pricing private insurance since the beginning of insurance and it is a prerequisite of conducting insurance freely. |
| **AIDA (acronym)**[32] | Artificial Intelligence or Data Analytics, defined as technologies that assist or replace human decision making. |
| **AIDA systems** | An AIDA system contains one or more AIDA models that work together to achieve a set of objectives. The decisions made by an AIDA system are "AIDA driven decisions". AIDA systems encompass not just the AIDA model(s), but also other parts of the decision making process that are required to get to the final AIDA driven decision, such as:<br>• Human contributions that impact decisions (humans might give input or use the model output to inform their final decision).<br>• Other governance or rules that affect decisions.<br>• Actions taken based on the system's output.<br>Monitoring and review mechanisms are also part of the end-to-end AIDA system and will impact AIDA driven decisions over time as feedback from monitoring and review is used to update elements such as the model, business rules and human inputs. |
| **AIDA firms**[33] | Companies that use AIDA to provide financial products and services. |
| **AIDA driven decision making**[34] | The process of making decisions using AIDA. |

| | |
|---|---|
| **AIDA driven decisions**[35] | Decisions and decision outcomes made or supported by AIDA. |
| **Behavioural fairness** | Rewards are offered or penalties imposed based on behavioural choices within the control of individuals.[36] |
| **Bias** | An inclination of prejudice towards or against a person, group of persons, object, or position.[37] It is a disproportionate weight in favour of or against an idea or thing, usually in a way that is closed minded, prejudicial, or unfair. Biases can be innate or learned.[38] Bias in AIDA systems can be a function of the data used to train the AIDA system, or of the AIDA system itself. |
| **Compliance** | The degree to which a person, product, or organisation obeys applicable laws, regulations, codes of conduct, or other legal and ethical norms. |
| **Confusion matrix** | Used to describe performance of a classification model or classifier. It is a table comparing true outcomes with the predicted outcomes. The number of correct and incorrect predictions are summarised with count values and broken down by each class. |
| **Data subject**[39] | An identifiable living person to whom a particular data item relates. In the context of this document, it refers to individuals and/or firms who are or may be affected by AIDA driven decisions. For example, a customer who applies for debt is the data subject for credit risk scoring use case. |
| **Drift bias** | A type of data bias that occurs due to:<br>**Data drift:** when the circumstances of an individual, or general population characteristics, change over time compared to the data a system is trained on, such as financial health, lifestyle, and occupation.<br>**Concept drift:** when the statistical properties of the target variable to be predicted change overtime, such as population mortality risk trends, technological advances, and evolving behaviour patterns, and which mean that relationships between factors and outcomes are constantly shifting. |
| **Ethics** | The work that is performed to satisfy a set of values, in accordance with principles, in support of governance. |
| **Explainability** | The ability to have explanations for decisions made (from an AIDA model and/or human process) that are understandable by an average adult; a higher standard than interpretability; a component of transparency (see also: “interpretability”, “model” and “transparency”). |
| **Fairness** | Related to the concept of justice; what is right, impartial and equal, without favoritism or discrimination; often assessed through the evaluation of decision outcomes, along with the process followed to reach the decision.[40] What is fair can mean different things in different contexts to different people.[41] Fairness is a guiding principle that aims to prevent the uneven distribution of harms. Expressing questions of fairness as mathematical problems is referred to as “quantitative measures of fairness”. These measures tend to be formulated as metrics that assess a criteria or objective, such as equal or equitable allocation, representation, or error rates, for a particular task or problem (see also: “bias”, “justice”).[42] |
| **Feature engineering** | The process of transforming raw data into features that may be useful as input to the objectives of a predictive model, based on domain/industry knowledge. |
| **Framework** | A prescribed approach and set of considerations. |

| | |
|---|---|
| **Governance** | Structures, processes, and mechanisms for oversight designed to assess whether norms are satisfied in a particular case. |
| **Historical bias / negative legacy bias** | When bias, human or otherwise, present in prior decisions is reflected in training data. Without adjustments, these biases will be perpetuated in new decisions from AIDA systems. |
| **Human rights** | Refers to the 30 Rights described in the Universal Declaration of Human Rights as proclaimed by the United Nations General Assembly in Paris 10 Dec 1948. |
| **Informed consent** | A decision of permission based on clear, accurate and relevant information about a situation where the implications of permission are adequately understood and made voluntarily. |
| **Interpretability** | The degree to which a trained professional can explain how a model arrived at a determination; a lower standard than explainability; a component of transparency (see also: "explainability", "model", "transparency"). |
| **Justice** | Underpinned by the notion of equal worth and the political standing of people. Similarly, it is the notion that people receive what they deserve. Deservedness is often defined by the normative values within a community, which could be an organisation, government, or other authoritative body. Adherence to justice in any instance can be measured when normative values of justice are named and weighted (see also: "fairness"). |
| **Justified** | When an action has a reasonable and defendable explanation. |
| **Measurement bias** | A type of data bias that occurs when there is systematic or non-random error in the collection of data and can occur on the input variables and target labels on which AIDA operates, thereby making the data more accurate for some people/situations compared to others. |
| **Model**[43] | In the context of data science, any AIDA algorithms, and algorithmic processes, with well-defined inputs and outputs, used to enable AIDA driven decision making. |
| **Model validation** | In the context of data science, the process of evaluating model performance on a test/holdout dataset (which should be separate from training data and not used during model training). |
| **Norm/ Normative**[44] | Norms can describe what it is socially agreed in a community to be ordinary or typical. Norms can also be ethical; prescribing what ought to or should be done. Often, what is ethical is different from what is currently being done (descriptive). Related to the FEAT Principles, norms offer guidance for what ought to be done – they are aspirational. For example, to avoid perpetuating discriminatory practices and unjustified inequalities, justice oriented considerations ought to be incorporated in ways they may currently not be employed. Used in this way, the term 'normative content' refers to guidance on what "ought" or "should" be done. It is about developing well-justified concepts, principles, commitments, and specifications for what individuals, groups, and organisations should do, rather than merely describing what they currently do.<br><br>Normative content (in the ethical, prescriptive sense) is ultimately grounded on values – both social and organisational. |
| **Objective** | Refers to the specific steps a company will take to achieve a desired result, as well as any constraints. The result is the goal.[45] |

| | |
|---|---|
| **Personal attribute** | Personal attributes are features that should not be used as the basis for decisions without reasonable justification. They are defined by FSIs in the context of each specific use case, and at a minimum cover any personal data included in the AIDA system, as defined in relevant data protection and anti-discrimination laws, and can also include non-personal data. |
| **Preprocessing bias** | Type of bias that can occur during data pre-processing in model development, when an operation (such as a missing value treatment, data cleansing, outlier treatment, encoding, scaling, data transformations for unstructured data among others) causes or contributes to systematic disadvantage. |
| **Principles** | The norms that describe how to implement values and set guardrails for what ought and ought not be done to operationalise values. |
| **Proxy bias** | A type of data bias that occurs in cases where input variables or target labels cannot be generated directly but are estimated using a proxy attribute. These proxies are either known/purposeful or unknown/accidental. Proxy data is used where it has the potential to provide better information than observed data or to fill in gaps where historical data may not exist, such as the use of a medical claims history to estimate future mortality risk due to the relatively low volume of death claims compared to hospitalisation claims. Unjustified used of proxies may adversely impact or marginalise certain population groups. |
| **Recourse** | The ability of a data subject to seek assistance when an outcome from an AIDA system is anomalous, negative, or unwarranted (see also, "redress"). |
| **Redress** | The act of rectifying the anomalous, negative, or unwarranted outcome from an AIDA system that has been learned about from a data subject seeking recourse (see also, "recourse"). |
| **Reproducibility** | The act of reproducing a model and its outcomes from scratch. Reproducible AIDA systems can be difficult to implement, requiring careful versioning of data, models, code, infrastructure, and even the random seeds used. |
| **Representation bias** | A type of data bias that occurs when groups, often minorities, are underrepresented in the data, affecting model training effectiveness. This could be due to selection bias during data collection, or simply the fact that the group is small portion of the overall population; this bias can lead to inadequate learning by the model and result in poor performance for certain groups underrepresented in the data. |
| **Standards** | An established norm or requirement; frequently aligned to a specific industry or repeatable task. Standards are often communicated in a formal document that establishes uniform criteria, methods, processes, and best practices. |
| **Transparency** | Visibility into the actions a system or organisation takes, and the decision making processes behind them.<br>• Where such visibility is aimed at data subjects impacted by an AIDA system's decision, it is referred to as external transparency.<br>• Where such visibility is aimed at internal stakeholders and to the organisation's regulators, it is referred to as internal transparency. |
| **Trust** | The belief that another party's goals are aligned with yours (even if the goals are divergent, they are thematically aligned), there is competence to achieve the goals, and one or both parties will work toward the goals. |
| **Values** | A set of ideals that describe what an organisation cares about and wants to promote and protect. |

# 10 Appendix C

## Global Regulatory Landscape on the Regulation of AI

### 1 Overview of Global Regulatory Developments

#### 1.1 Key Regulatory Developments

There are regulatory developments on the responsible use of AI in various jurisdictions globally – including in Singapore, Hong Kong, Australia, the United Kingdom, and the European Union. Not all of these are specific to financial services but are worth noting as they help provide context on regulatory and societal expectations on the responsible use of AI.



**Singapore**

- The Monetary Authority of Singapore published a set of 14 principles on Fairness, Ethics, Accountability and Transparency in November 2018. MAS aims to provide guidance to firms offering financial products and services (including banks and insurers) on the responsible use of AI and data analytics to strengthen their internal governance around data management and use, and to promote public confidence in the use of AI and data analytics.
- In 2020, three further initiatives were published in Singapore:
  - Model AI Governance Framework.[46]
  - Implementation and Self-Assessment Guide for Organisations.[47]
  - Compendium of Use Cases.[48]

  Together, these guidelines promote public understanding and trust by explaining how AI systems work, help organisations assess the alignment of their AI governance practices with the Model Framework, and provide use cases to demonstrate how local and international organisations of different sizes and across different sectors are implementing sound AI governance practices.



**Hong Kong**

- The Changing Landscape in Compliance and Supervision report,[49] published in August 2020, aims to promote understanding of the broad implications of AI adoption in the banking industry, and how to promote compliance and supervision by accounting for AI model risks such as data inputs, model design and validation.



### United Kingdom

- In June 2019, the Financial Conduct Authority (FCA) collaborated with the Bank of England to publish the Bank of England Future of Finance Report.[50] The report aims to monitor developments in the use of machine learning, develop principles and share best practices for responsible uses of AI, as well as explore the intersection of AI with current rules and identify where old rules need updating.



### United States

- Five financial regulators[51] recently issued a request for information, asking whether any clarifications from these regulators would support FSIs in the responsible use of AI, in compliance with existing laws and regulations.
- In August 2020, the US National Association of Insurance Commissioners published AI Guiding Principles,[52] which emphasise the importance of complying with existing regulatory obligations when using AI.



### The European Union

- In April 2021, the European Commission proposed the first regulatory framework on AI: the Proposal for a Regulation on a European approach for Artificial Intelligence.[53] The framework adopts a proportional and risk based approach, where AI uses will be classified according to risk levels (i.e., "minimal risks, limited risks, high risks and unacceptable risks"). The proposal remains subject to extensive review and consultation before possible approval by the European Parliament and the Member States.

## 1.2 Key Regulatory Trends in the Financial Sector

Some key trends across key global jurisdictions are highlighted in this appendix, which is intended only as a brief summary of the major developments.

FSIs operate in a highly regulated industry with robust requirements covering outsourced technology use, risk management, and governance. Because of this, regulators are looking to build on existing financial sector regulations, rather than add new requirements that are specific to AI. Regulators have devoted their efforts to increasing the industry's understanding of how existing controls and governance can be leveraged to implement and demonstrate Responsible AI practices.

As highlighted in the UK Committee on Standards in Public Life Artificial Intelligence and Public Standards Report (February 2020),[54] AI should not necessitate a fundamental reworking of practice. The focus should be on clarifying and revising current laws and standards so they can be more clearly applied to circumstances involving AI, rather than on establishing new regulatory bodies and laws.

The most notable trend for regulators in Asia has been to issue non-binding guidelines and principles on the adoption of AI; endorsing a principles based, technology neutral approach to the responsible use of AI. This regulatory approach can be seen most clearly in Singapore and Hong Kong, which is similar to the approach taken outside Asia by jurisdictions like the UK:

- **Singapore.** In addition to the FEAT Principles mentioned in section 2.4 above, the Infocomm Media Development Authority and the Personal Data Protection Commission have jointly published the Model AI Governance Framework, which sets out practical recommendations that are sector and technology agnostic. These recommendations could be readily adopted by organisations to implement AI responsibly.
- **Hong Kong.** In November 2019, the Hong Kong Monetary Authority published its High-Level Principles on Artificial Intelligence ("Circular").[55] Banks are expected to adopt a principles based approach when designing and adopting AI and big data analytics applications. Such principles include, amongst others: (a) using data of good quality, (b) ensuring an appropriate level of explainability of AI applications, (c) being ethical, fair and transparent, and (d) having board and senior management be accountable for the outcome of AI applications and decisions. Banks may apply the principles in a proportionate manner, depending on the nature of their applications and level of risks involved.
- **United Kingdom.** In June 2019, the Bank of England and the Financial Conduct Authority jointly published, the Future of Finance Report, which encourages the implementation of data standards and protocols to promote the responsible use of machine learning and AI by financial service providers.[56]



# 2 Assessing the Regulatory Implications of AI Projects in Financial Services

Different legal and regulatory requirements will apply in the different contexts in which FSIs are using AI, and there is no one-size-fits all approach. Regulatory issues should therefore be assessed based on the specific use of AI in question, and the jurisdiction and industry in which the system is being deployed.

There are five important considerations for FSIs to emerge from the work of Veritas consortium members. Table 2.1 summarises these learnings in the form of key considerations for FSIs when assessing the regulatory implications of their development and deployment of AI solutions.

| | |
|---|---|
| 1 | Adopt a materiality based approach to assess regulatory considerations when AI is being used to help focus efforts on more sensitive use cases. |
| 2 | Leverage existing internal risk management and governance frameworks when assessing the regulatory implications of using AI in specific use cases. |
| 3 | Collaborate with technology partners to promote transparency and accountability when deploying AI solutions. |
| 4 | Bring a diverse set of perspectives to discussions on regulatory issues relating to responsible AI to ensure ethics principles are appropriately applied, and that potential concerns are identified and dealt with early. |
| 5 | Share use cases and examples where possible to strengthen understanding across the industry of what regulatory issues may apply when using AI |

*Table 2.1: Key considerations for FSIs when assessing regulatory implications of AI*

## 2.1 Adopting a Materiality Based Approach to Assessing Regulatory Considerations

Assessing the materiality of use cases may be an important first step for FSIs to consider. A materiality based approach helps FSIs determine the proportionate application of governance and controls needed to implement Responsible AI principles, and ultimately focus efforts on more sensitive use cases. This can be contrasted to approaches that do not distinguish between the uses of AI and apply the same internal processes to all uses regardless of risk.

Given that AI is becoming embedded within all forms of software, it is increasingly pertinent for FSIs to adopt a materiality based approach to avoid undertaking detailed considerations of AI guidelines for every piece of software used. For instance, an FSI's use of AI systems that improve spell checking or recommend ideal meeting times is not likely to warrant the same level of scrutiny as other AI applications, such as facial recognition or the use of AI in decisions around credit worthiness or insurability. It may be useful for FSIs to apply a baseline set of Responsible AI criteria to any AI system being built, with a more intensive review procedure in place for sensitive uses where there is greater risk of harm.[57]

The materiality based approach has been endorsed by MAS in the publication of its FEAT Principles, where firms are encouraged to calibrate actions and requirements under their internal governance framework based on the materiality of AIDA driven decisions. Some of the suggested considerations to determine materiality under these principles include the extent to which the AIDA is used in decision making, the extent of automation of the process, and the severity and probability of impact on different stakeholders.

The materiality based approach can also be seen in the European Commission's Proposal for a Regulation on a European Approach for Artificial Intelligence.[58] While the consultation process is expected to result in refinements to the proposal, in particular to ensure that it can be implemented in a pragmatic manner, it is important to note that the proposal adopts a proportional and risk based approach. AI use cases are classified according to risk levels. AI systems of "high risk" (e.g., remote biometric identification systems) are a key focus in the proposed regulation and will be subject to greater scrutiny, carrying a set of five obligations. The Commission also identified "unacceptable" AI systems, for which a ban is proposed as they are considered contrary to fundamental EU rights. Examples are practices that:

- Have a significant potential to manipulate persons through subliminal techniques beyond their consciousness.
- Exploit the vulnerabilities of specific vulnerable groups such as children or persons with disabilities in order to materially distort their behaviour in a manner that is likely to cause them or another person psychological or physical harm.
- Are facilitated by AI systems which are manipulative or exploitative and affect adults.

To illustrate the operation of the materiality based approach, take the example of using AI for regulatory compliance relating to Know Your Customer (KYC) checks. This may be deemed a more material use case than using AI to verify wet ink signatures.[59] KYC checks are determinative of whether financial products/services are offered to the data subject (e.g., whether a loan or line of credit is approved). Outcomes from the KYC checks would have direct effects on the individual, as well as commercial implications, and regulatory compliance implications. As such, it may be necessary to have greater scrutiny of AI principles, such as fairness and internal accountability, to ensure any risks raised using AI are mitigated appropriately.

## 2.2 Leverage Existing Regulatory and Governance Frameworks

Existing risk management and governance frameworks are a useful starting point for FSIs to consider when implementing measures for the responsible use of AI. FSIs already operate in a highly regulated industry. Most jurisdictions already have robust regulatory requirements in place covering outsourced technology use, risk management, and governance. These regulations are almost always technology neutral, meaning that they apply equally regardless whether AI is used or not.

A key task for assessing the regulatory implications when implementing AI solutions would therefore be to identify the regulatory issues in a specific use case. Table 2.2 in Appendix C gives a hypothetical example of working through some of the relevant regulatory questions involved in an AI context, noting relevant Singapore regulatory requirements that may apply.

The alternative path to leveraging existing frameworks is to take the technology specific approach. The UK AI Act is the most obvious example where a jurisdiction has decided to carve out a certain set of technologies as "AI" and subject any use of these specific technologies to regulatory requirements that do not apply if the same use case was implemented in other technologies. It remains to be seen how effective this approach will be in practice.

| Phase | Some regulatory considerations | Jurisdictional examples |
|---|---|---|
| **Phase 1: input of data into AI technology** | • Data protection laws.[60]<br>• Confidentiality laws.<br>• Guidelines on outsourcing.[61] | While Singapore's Personal Data Protection Act (No. 26 of 2012)[62] does not have a specific provision relating to the use of AI, AIDA applications' reliance on internal and/or external data means that sector agnostic compliance requirements such as data protection regulation remains relevant. FSIs must therefore comply with the PDPA's framework on the collection, use and disclosure of personal data by private sector organisations in Singapore to protect the personal data of individuals, support public trust in the digital economy and enable innovation in the data space. |
| **Phase 2: processing of data by AI** | • Technology laws.<br>• Cybersecurity Laws.<br>• Risk management practices. | MAS has published several risk management practice guidelines, such as the Guidelines on Risk Management Practices for Insurance Business - Core Activities,[63] the Guidelines on Risk Management Practices - Technology Risk, and the Guidelines on Risk Management Practices – Board and Senior Management.[64] MAS generally requires all FSIs in Singapore to implement and maintain risk management frameworks, and to regularly update and develop such frameworks when new risks are identified. With the increasing adoption of AI solutions by FSIs, it is important for FSIs to evaluate the risks that the use of AI solutions may pose to their businesses. Thereafter, FSIs should update their existing risk management frameworks to include additional measures to address such new risks. |
| **Phase 3: output generated by the AI use case** | • Consumer protection laws.<br>• Anti-discrimination laws. | Singapore's Guidelines on the Provision of Digital Advisory Services[65] applies to FSIs offering or seeking to offer digital advisory services in Singapore. Digital advisors must put in place adequate governance and supervisory arrangements to effectively mitigate against any additional risks created by digital advice (e.g., that a fault or bias in an algorithm could adversely affect a large number or all of the digital advisors' clients who relied on the recommendations generated by the algorithms). |

*Table 2.2: Hypothetical example of regulatory considerations in Singapore in an AI use case*

## 2.3 Collaborate with Technology Partners

A growing area of interest is in the complementary roles of FSIs deploying AI technologies, and software developers providing AI solutions that FSIs might customise or deploy "off-the-shelf". By sharing technical and non-technical information about the capabilities and limitations of AI software being developed, technology partners can help an FSI assess the best way of deploying that software responsibly. This allows technology partners to work effectively with FSIs to promote transparency and ensure there is a degree of accountability for the software being developed. Technology partners may consider publishing or sharing information that fills any gaps between marketing and technical documentation. Customers will then be equipped with key information to deploy the AI products responsibly.

Regulators are also increasingly encouraging collaboration between FSIs and technology partners to improve understanding of how existing internal risk management and governance frameworks can be leveraged to implement and demonstrate good practices of responsible AI. For instance, the Veritas consortium has brought together 27 members, comprising MAS, financial institutions and technology companies to create a framework for financial institutions and technology company to promote the responsible adoption of AIDA. Similarly, the Ethics Guideline for Trustworthy AI[66] published by the European Commission in April 2019 was prepared by members of the High Level Expert Group on AI. This comprised members from prominent technology/financial companies like AXA, IBM and Bosch, as well as universities, and government advisers.



## 2.4 Ensure a Diversity of Perspectives

A diverse set of perspectives is of upmost importance in discussing regulatory issues and helps to ensure that the Responsible AI principles are appropriately applied, and that potential concerns are identified and dealt with early.

Diversity can come in the form of collaboration with a range of stakeholders (e.g., governments, regulators, industry associations, technology developers, etc. The Veritas consortium, as well as the EU's Ethics Guideline for Trustworthy AI[67] are examples of bodies that bring together diverse stakeholders. The cocreation approach between regulators and industry promotes transparency and accountability and ensures that principles can be translated into practical and implementable actions.

Diversity can also come from engagement with different teams within an organisation itself, including the teams responsible for developing the AI solutions, as well as the legal, risk and compliance teams. There should also be diversity within each team in terms of gender, ethnic groups, disability, race, etc. This approach ensures a diverse set of views contributes to decisions around the implementation of AI and ultimately ensures such decisions are reflective of the communities that organisations serve and operate within.

## 2.5 Share Use Cases With Industry

Given that it is largely the context in which an AIDA system is being used that triggers regulatory implications (i.e., not the use of system alone) sharing use cases and examples can strengthen understanding across the industry of what regulatory issues may apply. Initiatives like Veritas are valuable in creating opportunities to share use cases in this way and as part of Veritas Phase 1, use case illustrative document covering Fairness assessment in customer marketing and credit scoring was published.

The importance of use case sharing was in evidence in the multistakeholder collaboration between Microsoft, Standard Chartered, Deutsche Bank, Visa and Linklaters[68] to test the application of the FEAT Principles. Through this project, these industry participants worked collectively to assess the FEAT considerations involved in three different financial services uses cases: predicting travel patterns based on previous spending; automating the verification of wet ink signatures; and regulatory compliance related to KYC checks. The project partners found specific Principles have different implications in different contexts, and some principles will be more or less relevant depending on the use case.

Similar projects are being pursued outside the financial services industry through efforts to gather and publish use cases for responsible AI in Asia. This includes the pilot of Australia's AI ethics principles[69], and Singapore's two volumes of use cases, a number of which relate to financial services use cases.

# 3 The Way Forward for Veritas

With the rapid increase in deployment of AIDA systems by FSIs in Asia, a key experience in implementing guidance like MAS' FEAT Principles is that such regulator guidance provides a baseline set of Responsible AI criteria, which helps FSIs ask the right questions when deploying AI solutions. Guidelines on the responsible use of AI should acknowledge the importance of context in translating high level principles into practice. An important learning from discussions in Veritas and beyond is that each Principle will have different implications depending on the context, and a materiality assessment will be important to determine which Principle is relevant and how risks can be mitigated.

To this end, Veritas represents an important platform to support increased engagement across awareness of the issues raised by AI deployments across varying contexts, while also supporting efforts towards coherence in the guidelines and other requirements issued by regulators. Such an approach echoes a recent call by the Institute of International Finance (IIF) for a collaborative approach between regulators and industry to monitor and manage any risks associated with the use of AI.

Based on efforts to date through Veritas and other groups, there is clear potential for greater alignment around AI practices and the criteria used for assessing the materiality of AI deployments. Given the importance of context, rather than mandating a fixed or standardised approach, future efforts in Veritas could aim to develop indicative criteria for FSIs to draw on when assessing materiality in AI deployments.

Looking ahead, it is critical to build on existing work, encourage greater sharing of use case examples to improve understanding of how FSIs are implementing responsible AI, and to examine what regulatory questions FSIs need to address. This approach will strengthen the overall understanding of the use of AI across the financial services industry and instill confidence in regulators and consumers that these issues are being addressed.

# Annexure

## List of Veritas Consortium Members of Phase Two

















































































## LEGAL NOTICE

This report is prepared and issued by the MAS, AXA, HSBC, Microsoft, Standard Chartered, Swiss Re, UOB, Accenture, and TruEra.

All intellectual property rights in or associated with this report remain vested in the MAS, AXA, HSBC, Microsoft, Standard Chartered, Swiss Re, UOB, Accenture, and TruEra and/or their licensors. This report and its contents are not intended as legal, regulatory, financial, investment, business, or tax advice, and should not be acted on as such.

Whilst care and attention has been exercised in the preparation of this report, MAS, AXA, HSBC, Microsoft, Standard Chartered, Swiss Re, UOB, Accenture, and TruEra do not accept responsibility for any inaccuracy or error in, or any inaction or action taken in reliance on, the information contained or referenced in this report.

This report is provided as is without representation or warranty of any kind. All representations or warranties whether express or implied by statute, law or otherwise are hereby disclaimed.